

1 AI and Finance

DOI: [10.4324/9781003669357-2](https://doi.org/10.4324/9781003669357-2)

1.1 Introduction

Globalization boosts financial integration¹ resulting in higher resistance to potential shocks and refining the financial system. Moreover, globalization reduces trade barriers, increases technological innovation, stipulates better education, and generates an advantageous macroeconomic environment. Thus, financial reforms and assistance in technological innovation, particularly financial technology, are critical for promoting financial development, resulting in a more resourceful allocation of financial assets. The datafication of finance stimulates the financialization of the new economy, as financial transactions connect market participants around the globe and data is traded within and outside the financial system.

Financial contagion and spillover outcomes are asymmetric, where markets respond to positive and negative shocks in different ways and to different scales. Moreover, the integration of cryptocurrency and blockchain revolutionizes numerous industries, stimulating financial inclusion, transparency, and effectiveness in the advancing digital age.² It is characteristic that the financial industry offers the financial instruments that are needed in any business transaction, and managing the cost of financial instruments is diminished by the valuable use of technologies like blockchain and AI, which in turn slashes the cost of running corporations. Global financial markets have become gradually intertwined, and the SEC stated that regulators have to balance the aim of investor protection with the realism of the marketplace.³

So far, the power of AI technology seems to have been tremendously utilized by sellers, and pricing and data are central areas of policy in an AI-centric world. AI advances the capability to predict revenue outcomes of different policy options. Digitalization has caused faster and more effective service delivery by many corporations. Additionally, the consequences of

digitalization for building sustainable sociotechnical systems are crucial for new, large-scale technologies.

It is characteristic that globalization instigates amplified international competition, expansion of global markets and opportunities, along with altering surroundings and necessities for management practices. Hence, organizations desiring to stay competitive and progressive have to adjust to these transformations and employ new management tactics. In other words, rising international interaction, fluctuations in the technological environment via the fast advance of information technologies, and altering consumer requests generate new challenges for organizations in a globalized world. Thus, productive management in such circumstances necessitates new approaches guaranteeing flexibility, innovation, and effectiveness in the actions of organizations.

It has to be taken into account that the use of modern communication and collaboration technologies assists businesses in preserving operative interaction with global partners. Thus, corporations have to advance intercultural communication management capabilities and adjust to different cultural backgrounds to operate effectively in a global milieu. Economic issues such as variations in global markets, trade agreements, and currency fluctuations have a noteworthy influence on the financial condition and strategies of corporations. Moreover, alterations in consumer lifestyles mark the market state of affairs and the perception of corporations' products and services. It is worth stressing that climate change, energy competence, and environmental standards are becoming gradually significant for businesses in order to ensure their triumph in a global AI era. Hence, innovation and digital transformation are vital for corporate advancement via the employment of advanced technologies, operative use of data and analytics, and warranting flexibility and responsiveness to swiftly shifting market settings.

Auditing uncovers the mistakes and shortcomings of corporations and thus spots the rightful organization of production, along with the application of innovative technologies. Hence, an innovation audit is about understanding the successful organization of innovation and identifying which components are working productively and where extra efforts and adjustments are needed. Thus, the utilization of AI and AAI (Advanced Artificial Intelligence) systems will allow the employment of the best

factors in achieving the best outcome. Olga T. Cherney⁴ et al. argue that “It is also necessary to note five fundamental principles that must be observed when conducting an innovation audit: honesty, objectivity, professional competence and due care, confidentiality, and professional behavior.”

The current digitization of finance encompasses privately issued digital money and privately run digital payment systems; thus, these new technologies challenge the central balance of the sovereign public and private actors’ relative powers and roles in the financial system. Moreover, AI systems increase private financial institutions’ capability to over-generate risk and power in the financial system, without the parallel upsurge in the public’s capability to control that process; therefore, AI technologies make financial services more competent and accessible.⁵ Also, AI systems produce excessive generation and speculative misallocation of credit and money.⁶ Hence, an unstable financial system influences economic and social stability; thus, finance is a device for the geopolitical status of a country. Kenneth Krushel⁷ argues that

“The influence of AI and the all-encompassing technological age, challenges our pedagogy, challenges how we conceive our language, our imposition of geopolitical heuristics, and how we grasp dominion over the earth as well as our capacities for good and ill.”

1.2 Bitcoin as Money

It has to be considered that there is no need to distinguish the character of the transactions to assess a currency as a medium of exchange.⁸ A currency being a safe store of value is the key element for a means of being money, which is questionable at the moment, for digital/virtual currencies. Nakamoto⁹ was persuaded that the internet needed a currency that was, like it, decentralized, protected by cryptography, and favorable for peer-to-peer exchange. A group with pseudo-Libertarian political beliefs finds Bitcoin attractive owing to its lack of connection to any government, adhering openly to distrust of the world financial system, and the timing of Bitcoin’s

introduction, corresponding to the very bottom of the global financial crisis in 2008–2009.

Bitcoin¹⁰ is a non-fiat cryptographic electronic payment system that claims to be the world's first cryptocurrency. It is a peer-to-peer, client-based, entirely distributed currency that is not based on centralized issuing bodies being a “sovereign” to function. The value is produced by the users, and the procedure is executed by utilizing an open-source client installed on any computer running Windows, Mac, Linux,¹¹ etc. In Spain, the judgment of the Supreme Court¹² 20/06/2019 denied the recognition of Bitcoin as legal tender (money) but considered it an “incorporeal asset.”

Bitcoin is a homogeneous, virtual good that is entirely identical across all the online markets in which it is sold. The blockchain is a shared public ledger on which the entire Bitcoin network relies. It must be considered that network economics is a new field within the information society, and its principle is that products and services are produced and value is added via networks functioning on large or global scales. This is in contrast to industrial-era economies, in which ownership of physical or intellectual property originated from a single enterprise.¹³

On the one hand, the Court of Justice of the European Union ruled that Bitcoin was a virtual currency and that it could be accepted by the parties as a non-corporate means of payment.¹⁴ On the other hand, the Paysera judgment¹⁵ highlights that

“the definition of electronic money, which is considered to be any monetary value stored by electronic or magnetic means which represents a claim on the issuer, is issued on receipt of funds for the purpose of making payment transactions (...), and which is accepted by a natural or legal person other than the issuer of electronic money.”

To that extent, the Advocate General¹⁶ underlines that

“Bitcoin is not money as it does not fulfil the three functions of money: it is not a medium of exchange, nor a store of value,

nor a unit of account. These assets are not backed or managed by any central entity, nor is there any guarantee that they can be exchanged for cash. They are slow, costly and cannot be used to pay everyday bills. Its transactions are slow, costly, and cannot be used to pay everyday bills. It is not issued by any central public authority, and it does not have the same level of trust as an official currency such as the euro, which is backed by the central banks of the euro area.”

Hence, the Advocate General judges e-money to be Proton, miniCASH, Moneo, Geldkarte, and PayPal, and so, for the Advocate General, money is only stored in a network, the medium is “computerized” or virtual, and Bitcoin is not electronic money. Thus, the Advocate General’s view disagrees with the opinions articulated by the CJEU in the Hedqvist judgment, where it deems Bitcoin to be a virtual currency, which is legal money for the reason that it is understood to be within the autonomy of the parties and what they consider to have value, and is eventually the object of exchange between them.

Bitcoin is differentiated from fiat money since it does not have the agency costs of public money. John O. McGinnis¹⁷ argues that

“If Bitcoin is to grow into a more stable currency, it will need financial intermediaries to do so...Thus, even for a cryptocurrency like Bitcoin that should not itself be regulated, there is this paradox: to succeed, these structures currently seem to require financial intermediaries to function optimally and those intermediaries create the very agency problems that Bitcoin is designed to avoid.”

It is argued that Bitcoin does not entail faith in any public institution that creates money; on the other hand, it seems to be recognized that financial intermediaries are needed for its operation, which means going back to fiat money.

It is worth mentioning here that in order to flourish as a currency, Bitcoin and any other similar crypto require the same kind of financial mechanisms used to deepen the market for other financial assets, which means enabling price discovery and exchange-traded funds (“ETFs”). Also, while crypto was activated as a movement to free people’s financial businesses from the state, now the government is getting into the digital asset game with central bank digital currency (“CBDC”).

It has to be taken into account that the Bitcoin network operates as a public notary for all transactions conducted on the network, leaving a trace of what has transpired in the virtual universe, in cyberspace.¹⁸ It is worth mentioning here that cryptocurrencies are deemed to be volatile and lack trustworthy information, making them dangerous investments.¹⁹ The absence of regulatory certainty is a “massive barrier” to responsible innovation in the crypto ecosystem.²⁰ Furthermore, cryptocurrency is described as digital money based on blockchain technology. It is characteristic that Michael Conklin²¹ considers that “corporate investment in cryptocurrencies and the acceptance of cryptocurrencies as payment violates the stakeholder theory of corporate ethics...accepting cryptocurrency for payment and holding cryptocurrency as an investment clearly violates the stakeholder theory of corporate ethics.”

Payments and money transfers are the most obvious application of the distributed public ledger technology. Bitcoin transaction costs are much lower than those of traditional financial networks. The development of the “blockchain” is the innovative aspect of Bitcoin, which is a large decentralized database that has distinctive identifiers for each user and each Bitcoin. Moreover, virtual currencies and Bitcoin have the potential to revolutionize the means by which payments are processed, but only if they become ever-present.²² Could a broadly used virtual currency pose a risk to the stability of the financial system as a whole and to the economy more generally? It could be said that a virtual currency can be transformative if it is widely adopted, and a widely adopted virtual currency can threaten the financial system²³ as long as it is uncontrollable.

It is worth mentioning that it is very difficult to get accurate and up-to-date information about the number and volume of virtual currencies in circulation. The majority of virtual currency schemes have insignificant

market capitalizations of below €1 million and only 21 exceed this figure; only the top eight are above €10 million.²⁴ Bitcoin²⁵ is clearly the largest and most well-established virtual currency, with a market capitalization of more than \$28 billion.²⁶ On the other hand, Litecoin is scripted on an open-source global payment network that is never controlled centrally by any authority and uses a “script” as proof of ledger, which can be decrypted through a consumer-grade CPU. It could be said that Litecoin has a faster block generation rate compared to Bitcoin and thus presents a much faster transaction validation. Ethereum was launched in 2015 and is a decentralized software platform that permits Smart Contracts and Distributed Applications to establish and run without any downtime, control, impediment, fraud, control from any other third party. Zcash is a more secure, open-source, decentralized digital cryptocurrency launched at the end of 2016, and it is regarded as more secure because all transactions are recorded and registered on a blockchain with details about sending and receiving parties, and the sum transferred remains secure by presenting a provision for shielded transactions. There are also a number of post-Bitcoin virtual currencies that claim to be superior to Bitcoin, including Litecoin, GeistGeld, SolidCoin, BBQcoin, and PPCoin,²⁷ Dash (Darkcoin), Ripple (XRP), and Monero (XMR). It is worth mentioning that command regulation has a reputation for being inflexible, which means suffocating competition and inhibiting enterprise and growth. On the other hand, flexibility is a problem with regulatory technology, but being capable of delivering more precise regulation. Even though precision is desired in financial regulation as in any other field of law, it comes at the price of flexibility. Computer scientists must solve this problem by developing software that operates within flexible legal terms.

Under English law, cryptocurrencies are to be treated as property and so the UKJT accepted that cryptocurrencies have the characteristics of property, including being definable, capable of being owned and transferred, and having value.²⁸ Hence, English courts recognizing crypto assets as “property” means that they could be the subject of a proprietary injunction.²⁹ In *B2C2 v Quoine*,³⁰ the court accepted that Bitcoins could be the subject of a trust, and so were property. Nonetheless, in New Zealand,

cryptocurrencies were regarded as “digital assets,” a form of property capable of being held on trust.³¹ To that extent, A. Fowowe³² argues that

“While the US dollar has historically been the dominant currency in international trade, alternative currencies like cryptocurrencies and stablecoins are now more prevalent, opening up new opportunities for individuals and organisations to conduct cross-border transactions. However, using cryptocurrency also comes with legal difficulties, such as the issue of jurisdiction, whether cryptocurrency can be regarded as a property or being tangible under the Sale of Goods Act.”

It is vital that recognizing the inner workings of monetary systems generates valuable insights into the function and stability of traditional monetary systems³³ and of the value of the alternatives offered by digital networks such as Bitcoin³⁴ and in general crypto-assets. Hence, monetary exchange is more valuable at promoting cooperation than alternative record-keeping systems. The important factor is not the type of money, but rather the equality between units of money and the units of value in the economy, which means that the value of money has to be equal to the units of value of the total worth of trade in an economy. If the exchange rate is not determined by the demand and supply of money and traded goods and services, then instability produces problems in an economy. Price is the variable reflecting the exchangeability of any item versus money. Money has no role to play when individuals depend on shared knowledge of past conduct to imitate patterns of monetary exchange.³⁵ Bigoni et al.³⁶ argue that

“the differences in long-run efficiency, strategies and distribution of earnings in the Money and Memory treatments demonstrate that money performs a richer set of functions than just revealing past behaviors...the use of money imposes a discipline on what agents are willing to do in order to keep the

economy on a cooperative track. Such findings suggest that well-functioning monetary systems are not simply rudimentary arrangements for monitoring past conduct in society, but play a richer role than previously thought.”

The European economy rocketed once it was freed from a restrained money supply furthered by the development of new markets, new technology, and new payment systems that made possible long-distance commerce and movements of capital possible. Governments produce money themselves much more efficiently in the age of paper currency and electronic benefit transfers. Money is no longer tied to specie extraction.³⁷

Finally, in the developed world, government fiscal activity is subject to some degree of democratic accountability, and fiscal endeavor is a manifestation of democratic politics.³⁸ Governments are now much more energetically involved in control over the private production of money. Central banks now control the money supply by amending the price of money through the setting of target interest rates. Money production has become monetary policy, achieving the politically desired levels of growth, employment, and inflation. Thus, market, governmental, and technological changes signify that the effectiveness of private money creation stemming from fractional reserve banking is no longer so self-evident, even if its inherent instability is more and more obvious.

Banking law restricts entry into money augmentation.³⁹ Morgan Ricks⁴⁰ argues that

“private money often plays a central role in macroeconomic disasters, at least those emanating from the financial sector. The law of money encompasses the law of central banking... the law of money must also address the interface between public monetary institutions and private law.”

Banking regulation alters how capital is distributed both across sectors within an economy and across jurisdictions, escalating competition among governments that want to maximize national welfare.⁴¹

1.3 Regulating Virtual Currencies

Unlike the United States, the EU has employed a legal framework for the regulation of electronic money. The Electronic Money Institutions Directive 2009/110/EC⁴² outlines electronic money as electronically and magnetically stored monetary value and as epitomized by an entitlement on the issuer, which is issued on receipt of funds for the purpose of making payment transactions which means that the transaction is an act, instigated by the payer or by the payee, of placing, transferring, or withdrawing funds, regardless of any underlying obligations between the payer and the payee, and it is undertaken by a natural or legal person other than the electronic money issuer. Thus, it could be said that if a payment system accomplishes these prerequisites, then it is regarded as electronic money, and only electronic money institutions can issue electronic value. It must be considered that the vision behind this inflexible regulation is apparent, as what is taking place is the issuing of value into the economy. Therefore, Bitcoin would meet the legal definition to a certain degree, with the exception that it is not money that is issued in the consensus that is stipulated by the Directive. Due to the fact that there is no central issuing authority, it would be difficult to imagine how financial services authorities in charge of regulating EMIs could intervene in Bitcoin.

The ECB has expressed its view that a virtual currency like Bitcoin is “not money or currency from a legal perspective.”⁴³ Sweden defines currencies as being, inter alia, “tied to a central bank or a geographic area.” Bitcoin and other decentralized virtual currencies are designed to dispense with central bank and geographical localization, so they do not constitute “currencies” in Sweden. Many countries in the Eurozone have stated that virtual currencies do not constitute legal tender in their jurisdiction.⁴⁴ Moreover, FinCEN stated that “‘virtual’ currency is a medium of exchange that operates like a currency in some environments, but does not have all the attributes of real currency. In particular, virtual currency does not have legal tender status in any jurisdiction.”⁴⁵

The SEC does not have direct oversight of transactions in currencies or commodities, but the SEC is supervising the cryptocurrency-related activities of the market participants it regulates, including broker-dealers,

investment advisers, and trading platforms. Moreover, the SEC is allocating a noteworthy portion of its resources to the ICO market. In addition, the CFTC and SEC, along with other federal and state regulators and criminal authorities, will persist to work together to bring transparency and integrity to these markets, thereby discouraging and prosecuting fraud and abuse.⁴⁶ The SEC has issued investor alerts warning of the risks of cryptocurrencies and related products.⁴⁷ Moreover, the SEC has also issued investor alerts regarding the risks of original offerings of virtual currencies, and so, the SEC established a “Cyber Unit” within its Enforcement Division to pursue cyber-related misconduct, including violations of the securities laws encompassing distributed ledger or “blockchain” technology utilized by Bitcoin and initial coin offerings of virtual currencies.⁴⁸

Bitcoin is tied to existing political economic structures and markets and is therefore implicated in the creation of wealth and power inequalities at multiple scales due to the fact that Bitcoin and the governance of the system are controlled by an elite class of capitalists, coders, and some corporations.⁴⁹ Furthermore, the SEC’s rules for securities were found to apply even though the securities were purchased with Bitcoin⁵⁰ and as mentioned the CFTC has defined digital currencies as commodities.⁵¹ Likewise, the usual regulations applicable to those involved in the trading of commodities apply in the digital currency situation as well.⁵² In conjunction with the SEC, the Financial Industry Regulatory Authority (“FINRA”) released an investor alert concerning the risks of investing and transacting in Bitcoin.⁵³ The SEC publicly supported the IRS’s guidance of treating Bitcoin as property. Moreover, the SEC has become one of the federal agencies involved in enforcement actions and has pursued several Bitcoin companies for alleged fraudulent disclosures about business revenue and assets,⁵⁴ and for Ponzi schemes or fraud.⁵⁵ Moreover, the SEC ruled that the conduct perpetrated by the subject Bitcoin exchange constituted an investment in “securities,” and therefore, the federal district court has subject matter jurisdiction pursuant to Sections 20 and 22 of the Securities Act of 1933 and Sections 21 and 27 of the Securities Exchange Act of 1934.⁵⁶ The SEC⁵⁷ argued that the defendant’s Bitcoin product was a security for the reason that it constituted an “investment contract” under the language of the Securities Act’s definition of “security,” and an “investment

contract” was originally defined in SEC v. W.J. Howey Co, where the Supreme Court said that an “investment contract” is “a contract, transaction or scheme whereby a person [(1)] invests his money [(2)] in a common enterprise and [(3)] is led to expect profits [(4)] solely from the efforts of the promoter or a third party.”⁵⁸ Moreover, the SEC has treated Bitcoin as a security and applied antifraud and registration provisions of the securities laws, particularly Sections 5(a), 5(c), and 17(a) of the Securities Act of 1933, Section 10(b) of the Securities Exchange Act of 1934, and Rule 10b-5 of the Exchange Act.⁵⁹ While the SEC is only empowered with civil enforcement and administrative actions, it works closely with offering decisive information to the Department of Justice⁶⁰ to engage in criminal enforcement of the federal securities laws. For the reason that the SEC works closely with the CFTC, both agencies tackle options and futures in Bitcoin where regulatory overlap and coordination already occurs.⁶¹ Lately the SEC rejected an application by well-known investors to generate an exchange-traded fund tied to the price of Bitcoin, in large part because the markets where Bitcoin is presently traded are mostly unregulated and the SEC cited the lack of regulation as creating “concerns about the potential for fraudulent or manipulative acts and practices in this market.”⁶²

In other words, crypto assets are sometimes called virtual currency, coins, or tokens.⁶³ If cryptos are securities under the test in SEC v. Howey Co.,⁶⁴ they are sold in line with the federal securities laws that currently govern corporations, which means registration with the SEC and initial and ongoing public filings.⁶⁵

1.4 Monetary Policy and Financial Stability Policy

There is a monetary policy swapping between the pursuit of financial stability and the pursuit of price and output stability, which means that granting monetary policy another purpose produces confusion concerning the central bank’s commitment to price stability and, therefore, diminishes the nominal anchor with potentially undesirable consequences for economic outcomes. Safe assets play a central role in financial stability and monetary policy because of their significance for collateral production and their implications for macroeconomic stability.

Is there a relation between monetary policy and financial stability policy? Monetary policy and financial stability policy are associated with each other and consequently, there is no dichotomy between monetary and financial stability policy. Monetary policy influences financial stability, at the same time “macro prudential” policies upholding financial stability have an effect on monetary policy. Thus, if it is suggested that “macroprudential” policies are controlling credit bubbles by slowing credit growth and, therefore, slowing the growth of aggregate demand, then monetary policy needs to be easier in order to counterbalance weaker aggregate demand. Zekos considers that “macroprudential” policies are controlling credit bubbles by allowing rightful and not inflated credit growth, thus permitting the growth of aggregate demand on a real basis and values of the economy and not inflated ones.

Monetary policy and policies toward financial stability are two sides of the same coin.⁶⁶ The typical approach in monetary economics underlines the substance of managing market prospects, and this prospect conduit of monetary policy turns into the prevailing matter in many central banks, particularly among those that follow inflation targeting.⁶⁷ Milton Friedman and Anna J. Schwartz⁶⁸ argue that “leaving monetary and banking arrangements to the market would have produced a more satisfactory outcome than was actually achieved through governmental involvement.”

Instead, if policy rates are kept low to accelerate the economy, there is a greater risk that a credit bubble might occur, requiring tighter “macro prudential” policies to make certain that a credit bubble does not get on track. Therefore, the synchronization of monetary and “macro prudential” policies is essential in order for the three objectives of price stability, output stability, financial stability to be achieved. Frederic S. Mishkin⁶⁹ argues that

“financial sector plays a very prominent role in the macro economy and makes it highly nonlinear at times. This requires that we abandon the linear-quadratic framework for thinking about how to conduct monetary policy when there is a financial disruption. There is now a stronger case for a risk management framework that factors in tail risks that can

produce very adverse outcomes for the economy...there is a stronger case for monetary policy to lean against credit bubbles (but not asset-price bubbles per se), rather than just cleaning up after the bubble has burst. Using monetary policy to pursue financial stability goals is not an easy task, however, and research on how to monitor credit conditions so that it decisions to use monetary policy to restrict excessive risk are based on the correct information will be a high priority for research in the future. Finally, the financial crisis has made it clear that the interactions between the financial sector and the aggregate economy imply that monetary policy and financial stability policy are closely intertwined.”

Financial innovation and the new means to transfer credit risk lessen the informational substance of standard bank balance-sheet indicators. Bank risk surroundings need to be considered in conjunction with other indicators such as size, liquidity, and capitalization conventionally used to consider a bank’s capacity and readiness to supply new loans. Banks illustrated by lower projected default rates are able to put forward a larger amount of credit and to better protect their loan supply from monetary policy alterations.⁷⁰

The euro area banks play a key role as one of the central conduits for the execution of monetary policy, having a fundamental stance in the financial system. The function of banks in the execution mechanism is altered, primarily for the reason that the business of banks has undergone essential modifications in recent years, as a result of financial innovation, financial integration, and an upsurge in market funding.⁷¹

Ingredients of the banking area have moved away from the conventional “originate and hold” to an “originate and distribute” pattern of the banking company, which is much more dependent on market forces having an influence on the manner they award credit and respond to monetary policy desires. Borio and Zhu⁷² say that financial innovation, in correlation with alterations to the capital regulatory framework (Basel II), enhanced the effect of the perception, pricing, and management of risk on the methods of

banks. Also, Rajan⁷³ says that more market-based pricing and stronger dealings between banks and financial markets aggravate the motivation structures driving banks, leading to stronger relations between monetary policy and financial stability consequences. Likewise, W. Breuer and A. Knetsch⁷⁴ argue that

“Yet another breakthrough in digital finance is the emergence of decentralized finance, mainly characterized by cryptocurrencies and security tokens. The overall market capitalization for cryptocurrencies is around 1.19 trillion USD as of July 2023. These currencies act as an alternative to the conventional fiat currencies where consumers can utilize them for payments without the role of intermediaries, hence serving as a medium of exchange.”

It is worth mentioning here that due to the fact that the central bank is decisive in shaping monetary policy,⁷⁵ the increasing acceptance and use of cryptocurrencies are key development in the global financial landscape. The central bank ascertains the discount rates, which are the interest rates at which commercial banks borrow money from the central bank, thus shaping the interest rates charged by commercial banks to their customers and shaping the cost of borrowing for individuals and businesses. Consequently, employing cryptocurrencies will abolish the demand for financial and foreign exchange middlemen, reallocating currency supply away from central banks. Thus, this existing monetary structure was formed to affect inflation, economic growth, and financial stability by controlling the amount of money in circulation, which means that the uncontrolled production of means considered to be money in circulation will inflate the global financial system in due course according to the will of the elite controlling the whole cyberspace system and, consequently, the global economy owing to the advancement of the AI economy.

1.5 New Technologies and Dollar

Is there a need for the establishment of a new international monetary system encompassing electronic money? Have cryptocurrencies and crypto assets been influenced by conventional monetary policies? Will the dominance of the dollar be demolished due to the political failures of the United States in the global arena? The US hegemony over the international system has given its currency dominance over the global monetary system. Due to AI technology and the new political and economic reality of the global balance of power, a new global political and economic order has caused the global system's transformation. It has to be taken into account that the dollar's influence caused a financial crisis that affected European currencies via a decline in the value of their currencies expressed by a sharp increase in inflation rates and a fall in the competitiveness of their products. Moreover, the global system is experiencing a rapid end to the system of unequal exchange between Western countries and the rest of the world, along with an upsurge in the number of regional crises influencing the dollar's circulation. The China Securities and Regulatory Commission revealed that petroleum oil futures contracts commencing trading on March 26, 2022, will be denominated in Chinese Yuan rather than US dollars. Hence, de-dollarization is the process of dropping the use of the US dollar in international transactions and shifting to other currencies or payment systems.⁷⁶ Subsequently, the fluctuations in the dollar's value had effects on the global economy, and so the dollar exchange rate advances the competitiveness of foreign economies, principally US corporations, causing higher growth rates in these countries, entailing substitution effects and income effects. Moreover, there is a substitution or transfer consequence when fluctuations in the exchange rate influence the profit of a foreign sector in the international market at the expense of sectors in the dollar area. On the other hand, because of the transformation of the global economy and the rise of noteworthy global powers such as China,⁷⁷ the global economy is experiencing sub-regional trade, investment, and financial reform. To that extent, Osama Wagdi et al.⁷⁸ argue that

“The global oil market is ready to abandon the petrodollar; the weight of giving up the petrodollar will be more than 50% of the oil market; it will most likely take three to five years.

Russia and China are attempting to undermine U.S. control over the global monetary system for political reasons.”

Currently, US hegemony over the global monetary system means that the dollar will maintain extensive global dominance, which will be destabilized by the entrance of unrestrained cryptocurrencies and crypto assets in the global market, which will be handled by AAI systems that seem not to be controlled by the current elite unless the forthcoming AAI systems run by AAI humanoids are controlled behind the curtains by an AAI elite.

It could be said that China’s digital yuan sets a noteworthy challenge to the current dollar-dominated monetary order because the digital yuan serves not only China’s external objective of international digitalization but also another pole of countries opposing the dollar’s domination.⁷⁹ In other words, China’s digital yuan serves both its international objective of a greater role for the yuan in its mission to confront the dollar-dominated order and its national objective of greater state control over private money. Hence, it could be said that the digital yuan is not only a medium of exchange but also digital yuan as a digital currency confronts any private cryptocurrencies trespassing into the traditional domain of sovereigns. To that extent, Lan Cao⁸⁰ argues that “yuan internationalization and digitalization should be understood and considered within a wider context, possibly signaling the beginning of the end of dollar hegemony.”

1.6 Metaverse in Finance

AI and robotics technologies are amalgamated in several economic sectors such as financial trading and analysis. Algorithmic trading, driven by AAI algorithms has transformed market dynamics by permitting automated execution of trades. Hence, the integration of AI and finance has reformed the financial landscape, and by utilizing AAI, the speed, security, and efficiency of financial markets are improved. In line with this, blockchain technology, augmented reality (AR), and virtual reality (VR) have allowed the materialization of the metaverse⁸¹ being a virtual space where people interact comparably to the real world. To that extent, the metaverse is linked to the crypto-metaverse as a result of the strong integration of blockchain

technology and cryptocurrencies into virtual environments.⁸² In other words, the metaverse is a principal environment for the conception, trading, and ownership of digital assets and cryptocurrencies, generating the metaverse sphere for the global financial market. Thus, digital art, virtual real estate, and gaming tokens are acquired, sold, and traded within a digital terrain with the assistance of metaverse cryptocurrency coins and tokens. In Italy, tokens have been considered as “digital assets” under the provisions of Article 810 of the Italian Civil Code⁸³ and in France⁸⁴ tokens are regarded as “incorporeal assets.”

DeFi platforms are looking for opportunities within the metaverse, which means that users are involved in DeFi services like borrowing, lending, and gaining profit without leaving the virtual world. Blockchain technology and cryptocurrencies not only advance accessibility but also launch new ways for earning and employing cryptocurrencies. Thus, the metaverse alters the complexion of financial services which means that virtual banks, insurance companies, and investment platforms materialize within this digital world, presenting users with an alternative to conventional financial institutions. To that extent, smart contracts automate financial agreements, diminishing the requirement for intermediaries and intensifying the effectiveness of transactions. Thus, incorporating the metaverse into finance boosts global financial system accessibility by eliminating traditional barriers, inspiring innovation in financial services that lead to new products and solutions, and stimulating financial inclusion.⁸⁵

Fintech is transforming finance and contesting current regulations linked to financial inclusion. Moreover, virtual banking is distinguished from electronic banking owing to its absence of any reliance on physical infrastructure.⁸⁶ While virtual banks run exclusively online, traditional banks can offer more banking options in advancing market choice and, by this means stimulating financial inclusion without inevitably being digital. Hence, virtual banks, fintech start-ups, and online-based payment firms enter the supply side, advancing market selection and so, amplifying financial inclusion. Also, virtual banks boost financial inclusion by providing easier and faster bank account opening, easier modes to deliver identity for compliance with KYC requirements,⁸⁷ and algorithmically driven decision-making for loan approval. On the other hand, traditional

banks and virtual banks are subject to the same regulatory supervisory standards.⁸⁸

Interoperability among digital platforms⁸⁹ is a key attribute of the metaverse⁹⁰ and so the metaverse represents a prospective “parallel judicial universe, an independent meta-system with its own virtual jurisdiction.”⁹¹

It has to be taken into account that when a publicly traded company is attacked, the damage extends beyond the organization, causing financial losses to shareholders. To that extent, the appearance of the metaverse increases the attack surface for malicious players by generating more entry points for malware and data breaches. In line with this, by selling encrypted data back to its original owners, offenders obtain profits from personal data that had little value on the black market. Thus, digitization and the explosive growth of online business have caused cyberattacks, and corporations have become attractive targets. For the reason that corporations have centralized their data into information networks to augment competence, consumer data are more easily targeted in an attack.⁹²

1.7 AI, Decentralized Finance (DeFi), and NFTs

AI’s capability to produce distinctive content through algorithms displays a novel approach to NFT formation, where digital artworks are created autonomously or in collaboration with human artists. AI is not only an instrument for producing NFTs but also for managing them, which means that advanced algorithms investigate market trends, user preferences, and other data to propose NFTs to prospective buyers, thus augmenting the discovery process in marketplaces. Hence, AI competencies are seen as an instrumental asset for both creators seeking visibility and collectors looking for their next acquisition. In other words, AI’s function in NFTs allows confirming the authenticity and originality of digital assets, which means that AI aids in identifying prospective copyright infringements or duplicates and so warranting the integrity of the marketplace. Nonetheless, the employment of AI in the NFT space causes concerns regarding authorship and copyright, particularly when AI-generated artworks become tokenized as NFTs. Thus, while AI technology progresses, its impulsion on the NFT

ecosystem is increasing and restructuring the terrain of digital creativity and ownership.⁹³

It is characteristic that the success of new product development efforts is based on the creativity of the underlying ideas, which means that creativity encompasses the production of new ideas that are both original and valuable. Therefore, innovation comes as an implementation of ideas into new products and processes. Consequently, creativity is regarded as the first stage of the innovation process, followed by implementation. What is the role of AI in creativity? AI will be utilized in generating new innovations instead of humans due to its capacity, and so, AI creations have to be awarded the relative IPRs.

It has to be taken into account that the synthesis of Decentralized Finance (DeFi) and NFTs⁹⁴ denotes a considerable evolution in the digital economy, and so, this convergence presents innovative financial products and services, escalating the utility of NFTs in finance. DeFi platforms employ blockchain technology to generate financial systems that run without central authorities, such as banks or governments, presenting services like lending, borrowing, and investing directly on the blockchain. To that extent, NFTs are becoming fundamental to DeFi by representing distinctive assets utilized as collateral for loans or as part of investment strategies. In other words, an NFT art piece employed as collateral to secure a loan on a DeFi platform opens up new liquidity prospects for NFT holders, permitting them to leverage their digital assets without selling them. Hence, DeFi projects tokenize real-world assets as NFTs, making them tradable and functional within the DeFi ecosystem, and so, tokenizing property titles, luxury goods, or unique experiences, which means that NFTs expand involvement in DeFi, advancing more diverse investment opportunities. Nonetheless, amalgamating NFTs with DeFi brings forward complexities, such as evaluating the value of irreplaceable digital assets for collateral reasons and handling the risks linked with their volatility. Hence, the decentralized character of these platforms presents extra challenges in safeguarding security and averting fraud, and so, demanding constant innovation in design and platform governance. In line, Chen Jun and Danny Friedmann⁹⁵ argue that

“NFT platforms explicitly point out that if one buys a “digital collectible,” one will get the NFT in combination with copyright of the underlying work. Its monetization process is linked to fiat money (namely the Renminbi), but the tradability of NFTs is limited to trade within separate platforms/chains.”

1.8 NFTs and Digital Economy

As mentioned earlier, the advent of NFTs is affecting the digital economy by presenting novel notions of ownership, value, and exchange in digital spaces. To that extent, NFTs confront conventional economic models by allowing digital assets to be uniquely owned and traded, much like physical assets and, thus, obscuring the lines between the tangible and intangible in economic terms due to the fact that NFTs redefine ownership. It could be said that blockchain-based proof of ownership for digital items means that NFTs form scarcity and exceptionality where duplication was once inexhaustible. Hence, scarcity is becoming a central economic principle that NFTs apply to digital content, possibly escalating its value and making it a tradable commodity within the digital economy. Likewise, NFTs are accelerating the formation of new markets and opportunities for monetization, which means that artists, musicians, and creators can sell directly their digital works to a global audience, sidestepping established intermediaries and, thus, keeping a larger share of the profits. Thus, the AI economy not only energizes creators but also accelerates a more active and diverse marketplace for digital goods. In line with this, Joerg Osterrieder⁹⁶ et al. argue that

“the implications of NFTs for the digital economy are profound. They are catalyzing a shift towards a more decentralized and creator-centric marketplace, where digital assets are recognized and valued in new ways...t’s clear that they are not just a passing trend but a significant evolution in

the digital economy, with the potential to reshape how we create, share, and value digital content.”

It is worth noting that given that cryptocurrencies are the default money for NFT transactions, their price patterns expose this interconnection. Hence, NFTs are exchanged in cryptocurrency exchanges but largely perform as digital assets, holding value, empowering transactions, and operating as units of account. To that extent, exploiting NFTs to ascertain digital ownership is innovative and safe, thus inserting security into ownership claims and enhancing trustworthiness. Also, prior to NFTs’ appearance, there were no decentralized ways to quickly and definitively authenticate digital artwork ownership. It is argued that NFTs, predominantly in decentralized finance (DeFi) sectors, steer more as a result of investors embracing a collective mindset and following investment trends, resulting in a shortage of rational asset analysis, which means that NFTs are considered to be a different asset class from other cryptocurrencies.⁹⁷ Moreover, NFT price dynamics were tied to Bitcoin market movements and trends in its early phases. Suliman Alshahmy⁹⁸ et al. argue that “In today’s financial landscape, emerging digital assets’ independence from conventional financial instruments is important. This theory holds that cryptocurrency prices drive the NFT market.” It seems that NFT and cryptocurrency markets have acquired attention for their value due to the explicit causes of price variations and their consequences. Also, Bitcoin market efficiency and returns are time-varying, and so, Bitcoin investor attention concentrates on its links with returns, trading volume, volatility, and liquidity. Weiwei Guo⁹⁹ et al. find that “factors such as changing user composition and increased regulatory scrutiny alter the dynamics of on-chain factors and their influence on market efficiency.”

It is characteristic that buyers are expected to exploit the NFT as collateral to fund their purchase, rather than taking a traditional mortgage. R. Wilson Freyermuth¹⁰⁰ et al. argue that

“the NFT financing model is not a mere technological upgrade, but rather transports parties out of the domain of traditional mortgages and into secured transactions law, with

significant legal and policy implications. Most worryingly, it exposes borrowers to swift and irreversible home liquidations in cases of default, robbing them of the protections historically afforded to homeowners.”

Hence, these dealings are still real property acquisitions financed with a loan.

Finance has become one of the most globalized and digitized sectors of the economy. Moreover, globalization, digitization, and money are boosting AI in finance forward at an ever-increasing pace. In line, Zetzsche¹⁰¹ et al. argue that

“the use of AI in finance comes with three regulatory challenges: (1) AI increases information asymmetries regarding the capabilities and effects of algorithms between users, developers, regulators and consumers; (2) AI enhances data dependencies as different day’s data sources may alter operations, effects and impact; and (3) AI enhances interdependency, in that systems can interact with unexpected consequences, enhancing or diminishing effectiveness, impact and explainability. These issues are often summarized as the “black box” problem: no one understands how some AI operates or why it has done what it has done, rendering accountability impossible.”

It must be taken into account that the expanding uses of black box AI¹⁰² by governments and private corporations can have negative outcomes for people due to malfunctioning algorithms and so, affecting an entire economy, which means that it is imminent the development of AAI tools for controlling and correcting AI predictions and AI decision-making automatically. Brandon L. Garrett and Cynthia Rudin¹⁰³ argue that “Regulations and statutes regarding the deployment of AI in criminal law settings should directly require interpretable and validated AI.”

It could be said that bolstering cyberspace governance of regulated financial market participants through external regulation is the way forward in AI era. In other words, the most effective course forward implicates regulatory approaches which bring the human into the loop, augmenting internal governance through external regulation. Thus, the application of the principles of governance in an AI environment is an answer.

In the context of finance, the post-crisis concentration on personal and managerial responsibility systems offers a central external framework to augment internal accountability in the context of AI, which means that AI-tailored manager responsibility frameworks, augmented in some cases by independent AI review committees, as developments to the conventional three lines of defense, are the most effective way of dealing with AI-related matters not only in finance concerning “black box” problems but also possibly in any regulated business.

In the end, the use of AI for managerial tasks implies that AI technologies are a new set of agents in corporations. Moreover, AI systems have practically unlimited information-processing capability and present perfectly consistent behavior. However, they can introduce statistical biases due to unsuitable data; thus, AI and shortly AAI systems have to be utilized in order to correct any biased data instantly. In other words, current AI can be pre-programmed to engage in clear mathematically defined goals while taking into consideration legal and ethical constraints. On the other hand, AAI systems can objectively run and directly correct any algorithm by instantly choosing to insert of the newest unbiased data. Additionally, AI technology is projected to deliver welfare and security, which makes it a strategic resource for governments, meaning that AI should not be an elitist project handled by political-administrative and corporate interests.

1.9 Financing Startups

It has to be taken into account that financing new ventures is risky and characterized by asymmetric information and multiple incentive problems. In other words, most startups are not attractive borrowers. Venture capital backing substitutes for positive free cash flows in the context of startups and that intellectual property performs a central role function in substituting

tangible assets as collateral, making venture debt attractive to lenders. Moreover, early-stage venture capitalists choose venture debt if their portfolio corporations have low upside prospective and if they cannot gain from the value that a late-stage venture capitalist adds or if uncertainty is low. Consequently, a venture loan is comprised of two key elements: a loan and warrants.

Startups typically have no positive cash flow, little collateral to offer, and high bankruptcy rates, making them poor loan candidates. Nevertheless, venture loans, as a hybrid form financing, embrace both a loan and a warrant and are used in practice.

In other words, innovative startups have limited access to debt, not having positive cash flows (yet), have inadequate collateral assets, and are portrayed by uncertainty concerning the success of their business model and high bankruptcy rates. Also, venture loans are regarded as “hybrid form” of financing between straight debt and convertible bonds, which means that venture loans are frequently linked with the maturity of the startup, performance-oriented investor types, and startups with financially strong committed investors. To that extent, new ventures are subject to the liabilities of newness and smallness, meaning their bankruptcy rates are higher and their access to resources is limited compared to those of established corporations. In line with this, Nico Lehnertz et al.¹⁰⁴ argue that

“venture loans are associated with strongly committed existing investors, which stimulates the requirements of venture lenders and is signaled through large invested capital amounts per investor in previous rounds. Furthermore, we find that venture loans are associated with rather mature startups and offer empirical indication that the medical, health, and life science industry with clear milestones provides good conditions for venture loans.”

The warrant permits the venture lender to participate in the return of successful exits and so, in contrast to straight debt, a venture loan offers upside potential. Nonetheless, a venture loan has to be distinguished from convertible notes that are often offered by existing or future equity

investors. To that extent, the intellectual property of new ventures is advocated as a substitute for missing tangible assets as prospective loan collateral, and so, patents are imperative tangible assets offered as collateral for venture debt.¹⁰⁵ Thus, startups offering appropriate intellectual property as collateral and as quality certification are appropriate candidates for venture loans.¹⁰⁶ Moreover, with increasing maturity, startups gather tangible assets and become more profitable, leading to an increase in debt use.¹⁰⁷

The financial growth cycle of startups usually starts with one's own capital injections and support from family and friends, followed by participation from business investors. Thus, startups in intermediate or later stages of the lifecycle attract venture capital investors.

1.10 Finance Models

Increasing digitization and automation of processes in all parts of society have initiated a deliberation on whether people want to count on algorithms. Simultaneously, algorithms present openings for new business models relying less on human decision-making and human interaction, but more on pre-defined rules and processes. The financial industry offers many applications for algorithms in trading, asset management, and financial advice. Robo-advisors grounded in algorithms are utilized to help human fund managers or investment advisors. M. Germann and C. Merkle¹⁰⁸ argue that

“Participants acting as investors can tie their incentives to either a human fund manager or an investment algorithm. We find no sign of algorithm aversion: participants care about returns, but do not have strong preferences which financial intermediary obtains these returns. Contrary to what has been suggested, participants are neither quicker to lose confidence in the algorithm after seeing it err. However, we find that participants' inability to separate skill and luck when

evaluating intermediaries slows down their migration to the algorithm.”

As a result of economies of scale, algorithms present low-cost products and advice to retail clients serving many people at a time, while fees for human financial advice are often prohibitive for small investments. Financial algorithms now directly interact with retail investors, and new fintech corporations relying on technology enter the market for investments all over the world.¹⁰⁹

Financial technology is improving financial processes and activities predominantly through the use of computers and smartphones. Robo-advisory platforms use algorithms to present financial planning services without human supervision, thereby delivering asset management solutions that are both more transparent and less expensive.¹¹⁰ It could be said that the decentralization of finance has advanced the development of innovative financing formulas based on which corporations, entrepreneurs, or project developers raise funds in open calls through cyberspace, receiving cryptocurrencies in exchange for digital assets entitling buyers to an exclusive right, reward, or financial claim.¹¹¹ Yan Chen et al.¹¹² argue that

“decentralization has an inverted U-shaped relationship with platforms’ market capitalization, developer attention, and development activity. We further examine factors driving the decentralization of platform governance and find that digital platforms of the infrastructure layer—relative to those of the application layer—have a tendency to become more decentralized. This tendency, nevertheless, can be offset by experienced leaders to achieve semi-decentralization.”

Tokens representing ownership of an asset, such as corporate debt and equity, are called security tokens, and their purchase through an STO is considered an investment subject to securities laws.¹¹³ Contrasted with traditional financial products, the issuance of blockchain-based tokens through ICOs/IEOs/STOs presents enhanced public tradability,

transparency, low entry barriers, competence, automation, and high-cost efficacy being both alternatives and complements to other innovative sources of financing, such as crowdfunding and VC.¹¹⁴

Taking into account that Blockchain¹¹⁵ platforms are written in code, empowering data ownership and decision-making processes, blockchain facilitates groups of stakeholders to collectively produce and distribute value among themselves in new and more decentralized modes.

Business financing optimizes the mix of loans, bonds, common stock, and many other feasible facilities. Moreover, the amplexity of a particular mode of financing a business diverges depending on the precise situation of the corporation and the economy. New corporations are not able to acquire loans or issue debt.¹¹⁶ Moreover, information asymmetries over and above interest payments inadequate to compensate financiers for the fairly high possibility of losses are among the reasons which means that new corporations have to count on equity financing, encompassing the additional advantage of investors who support the management with their own experience and networks.

It is characteristic that in a centralized token economy,¹¹⁷ central players allow interoperability with other central players so that tokens are transferred between them across digital platforms via SWIFT. Moreover, technologically, asset ownership management counts on digital platforms controlled by central players and, therefore, controlling a platform has power over the stored data since the processes executed via the platform cannot be authenticated by other players. Also, the main players gather decision rights for the governance of provided digital platforms, dependencies on the central actors are augmented, and all individual players have to trust the central players. Likewise, players have to trust that the central player will not stop offering services on a digital platform or increase service fees.

On the other hand, a decentralized token economy tackles drawbacks linked with the involvement of central players in asset ownership management by cutting dependencies on central players and their digital platforms. Hence, various players cooperatively run a digital platform, dividing the linked duties, such as for platform upholding.¹¹⁸

The decentralized token economy empowers direct transactions between players without mediation by banks, which means improving cost competence and via tokens boost suppleness in managing asset ownership which means that multiple players can contribute to decision-making for the central missions in asset ownership management that are presently executed by central players. Correspondingly, DLT permits the automation of key assignments of the central players entangled in centralized token economy. Therefore, cross-ledger interoperability (CLI) advances the technical capabilities to connect isolated DLT systems prevailing over shortcomings, such as limited network consequences and suppleness in business relations, which means that technical standards empower digital support of governance mechanisms.

The goal of centralized exchanges (CEXes) is to offer a method to trade and custody currencies and cryptocurrencies. A decentralized exchange (DEX) offers a method to trade cryptocurrencies in a peer-to-peer manner using smart contracts, thereby reducing default or scam risks and adding privacy delivering full transparency. Nevertheless, a DEX presents the likelihood to trade most cryptocurrencies at any time via a system of automated market.¹¹⁹

Tokenization¹²⁰ permits trading via a distributed ledger, a variety of ownership and access claims, licenses, and royalty rights as an alternative to conventional, paper-based exchange. Also, tokenized claims concern both physical and digital assets, which means that tokenization offers investors a new mode to obtain “fractional ownership”¹²¹ and so, investors can swap tokenized claims for physical assets such as real estate, artworks, etc. In other words, “tokenization” concerns the holding and transfer of financial assets on decentralized blockchains or distributed ledgers.¹²² In line, Rosa M. Garcia-Teruel and Héctor Simón-Moreno¹²³ argue that “private law rules may be adapted to the tokenization of property rights, which may contribute to the establishment of a digital market for the trading of asset-backed tokens worldwide.”

Concerning tokenization, Darian M. Ibrahim¹²⁴ argues that

“crypto issuers would be required to include a short, prominent “warning box” on their websites that includes only

unexpected and harmful features of the crypto. Coupled with the whitepapers already provided by crypto developers—a shining example of voluntary disclosure working—the warning-box add-on completes the crypto regulation picture and properly balances innovation and investor protection.”

It has to be taken into account that legislation is illustrated as too weak and slow to protect individuals in a rapidly developing technological world due to the fact that changes in the technological setting are so quick and profound that regulations become quickly ineffectual at appropriately protect individuals.¹²⁵

Crowdfunding is the process of raising capital for a campaign, a project, or a new business through the rather small contributions of a fairly large group of individuals over a set period of time, permitting all sorts of promoters to find financing for all kinds of projects.¹²⁶ The crowdfunding is made possible by technology through online platforms offering its contributors new communication means and a model simplifying the capital-raising process.¹²⁷ Also, crowdfunding establishes a market connecting directly the funders with the promoters of projects, thus eliminating the requirement for intermediaries such as banks or venture capitalists. Of course, the online platform is required for obtaining funds comparatively simply, efficiently and quickly.¹²⁸ Crowdlending is known as peer-to-peer lending (P2P) or peer-to-business lending (P2B), and capital is obtained directly from the original source of funds rather than from a third party. It is characteristic that while P2Bs’ capability to directly connect investors and corporate borrowers relieves them from the requirement to hold costly regulatory capital, it subjects them to the impact of liquidity shocks. On the other hand, banks’ liquid inventories permit them to bridge funding gaps, but at the cost of subjecting them to credit risks and the resulting regulatory oversight, involving capital requirements. In other words, the variety in designs used in the P2B industry, along with the alterations that have taken place in Funding Circle, reveals that the financial industry is still in a stage of experimentation.

Crowdlending is a part of the FinTech “revolution” disintermediating financial services. It has to be taken into consideration that the P2P/P2B

online lending platforms have challenged traditional banks and intermediaries by presenting more cost-efficient offerings as a result of the scaling of the newly adopted technologies, and so, these services tender higher returns for lenders and lower interest rate costs for borrowers.¹²⁹ It could be said that P2P platforms are closely bank substitutes serving the same borrowers.¹³⁰ Also, the platforms do not take credit risks but rather decentralize these by spreading them to the crowdlenders. It could be said that peer-to-peer (P2P) lending substitutes bank loans, having negative influence on national bank loan balances when P2P lending balances is big.¹³¹

In other words, crowdfunding implicates the financing of projects by large numbers of people via digital platforms serving as electronic marketplaces, which means that peer-to-peer (P2P) lending is one of the two main methods of for-profit crowdfunding. The other method is equity crowdfunding, where early-stage ventures raise outside equity from the crowd via digital platforms. Additionally, P2P lending involves the crowd and financial institutions funding loans to small businesses or to consumers.¹³²

It has to be taken into account that market participants have articulated fear regarding the cost and complexity of relying on Regulation Crowdfunding encompassing disclosure regulations.¹³³ Crowdfunding presents an equivalent situation to ICOs in that it also implicates selling risky investments to unaccredited investors through general solicitation. ICOs present high levels of information asymmetry between developers and coin buyers, both in terms of a coin's newness and its inspiring blockchain technology. Nonetheless, crowdfunding disclosures have been inefficient and unread by most investors.¹³⁴ Thus, disclosures reduce information asymmetry ex ante and permit fraud enforcement ex post.¹³⁵

P2P marketplaces provide two novel sources of external debt finance; on one hand, P2P business loans offer medium-term finance for maturities of up to five years, and on the other, invoice finance presents short-term asset-backed loans with up to one-year maturity. Moreover, P2P or marketplace lending diverges from bank lending since it is funded via a cyberspace portal and is not being subject to Basel III and other formal banking regulations, constituting a unique method of outside entrepreneurial

finance.¹³⁶ In line with this, Jerry Coakley and Winifred Huang¹³⁷ argue that

“the success of the P2P business lending market stems from the symbiotic relationship between financial institutions and small investors as the lenders. The former exploits the collective wisdom of the crowd while the latter views large institutional investments as a certification effect that reduces information asymmetries and sometimes herds in their wake.”

It has to be taken into account that the fintech platforms have advanced the lending process by rendering the P2P lending and microloans that adjust to the borrower’s wishes. Moreover, different P2P lending platforms utilize the latest technology in credit risk assessment and lending suitability, allowing them to present fitting solutions.¹³⁸

Online lending platforms by directly connecting retail lenders with corporate or consumer borrowers have been believed to be a new model of credit finance: decentralized, transparent, disintermediated, and more information-efficient. On the other hand, Julian Franks et al.¹³⁹ argue that “the main problem of the platform was its vulnerability to liquidity shocks, resulting in sizable deviations from information efficiency. These increased over time due to a growing role played by non-crowd players, particularly large investors and algorithms.”

Has the right to print conventional money been utilized for conquering the globe by an elite? Will cryptocurrencies and tokenization be the new tools for expanding old elite colonization of the globe? Is the supposed decentralized crypto a veil in order to eventually control the printing and circulation of money according to the interests of the same old elite? Taking into account that any digital money facilitates fast, seamless, and around-the-clock transactions between consumers and corporations, the tokenization of the financial system will implicate a complementary relationship between digital public money, like CBDCs, and other forms of digital private money. CBDC is a device for central banks to maintain stability within the digital realm, essentially tackling the limitations of traditional forms of money.¹⁴⁰

It seems that tokenization is transforming how money better serves financial demands and, as a result, it generates new openings for consumers and corporations to transact in new spaces and new methods which means opening new paths for wider inter-industry settlements, assisting financial inclusion, empowering seamless cross-border transactions, and finally boosting the competence of capital markets.

It is characteristic that DeFi has made it possible to cut transaction costs, promote wider financial inclusion, enable open access, increase transparency, tolerate borderless finance, and generating new opportunities for entrepreneurs and innovators. Consequently, DeFi has advanced finance via a financial system without borders, jurisdiction, and centralized control, while raising significant challenges and threats to financial stability.¹⁴¹ Decentralized finance, or DeFi, replicates financial services with smart contracts over blockchains, and so DeFi operates as money “Legos”¹⁴² which means that cryptocurrencies are a tradable financial product representing a digital asset acting as a digital form of money.

1.11 Data and Finance

Advances in AI systems generate a variety of smart connected products and services, software, data analytics, and connectivity.¹⁴³ Big data allows for information handling, and so, predictive analytics tools that control data volume, velocity, and variety are utilized. Additionally, the explosion of digital data is critical, as by using big data, managers can measure and grasp more insight into their corporation and directly translate that knowledge into enhanced decision-making and performance.¹⁴⁴

AI has the ability to accurately interpret external data, to learn from such data, and to use those learnings to accomplish particular objectives and tasks through flexible adjustments. Additionally, AI concentrates on financial detection and intrusion detection systems in data networks through data mining techniques and neural networks.

Does AI monetize data? It has to be taken into consideration that monetizing collected data is a revenue source for large corporations, such as Facebook and Google. Huge data footprints are aggregated and ultimately exploited to cross-sell and upsell products to customers which means that

data is valuable. The scale of automation has enriched the industrial corporations with more intelligent and self-adaptive improvements are made through AI. AI software is creating unnoticed business prospects and societal values, with even smarter chat bots offering expert assistance and robot advisors conducting direct research in many fields.

It is argued that learning algorithms and big data, in combination with human creativity, assist organizations in optimizing their business and better predicting and controlling their processes and applications.¹⁴⁵ Hence, corporate information is collected, shared, and enriched via co-creative computational systems and assistive technologies that augment human capabilities and productivity. Are AAI humanoids the intermediary/intermediary between humans and machines? At the moment, AI systems are used by humans in order to advance any business process, but AAI humanoids will be able to control the entire economic process without any human involvement. It is supposed that AAI humanoids will not be the new deadly weapon in the hands of the old elite in order to govern the globe once again.

It has to be taken into consideration that data explosion is the central driver of technological innovation, and so, corporations in order to handle large volumes of data, exploit cloud where the participating corporations host their applications and data by utilizing machine learning algorithms.¹⁴⁶ It seems that new technologies generate new problems due to their function, and so, a new technology comes to solve the problem, which means that AI technology has to be effective in solving any malfunction of the technology and not merely a new technology required to address its application consequences and problems due to its function. On the other hand, AAI technology will deal with all the problems caused by its function and offers solution directly, not requiring a new technology to solve its misapplications.

1.12 Digital Financial Services

It has to be taken into consideration that the rapid growth of digital financial services (DFS) has had a transformative impact on the finance of corporations.¹⁴⁷ Julian Gruin¹⁴⁸ specifies that “the rise of Chinese fintech

represents more than a market-based utilization of new financial technologies in response to an entrenched state-dominated banking system, but rather constitutes an important institutional foundation in the ongoing construction of an authoritarian capitalist system.”

China has resisted the limitations of globalized finance, but it endorsed an acculturated set of strategies to both incorporate and constrain the power of global financial integration in order to avert macroeconomic risks. Additionally, the political economy of Chinese fintech encompasses many dynamics of change within the financial system, leading to the transformation of social and economic governance structures.¹⁴⁹

Financialization utilizes the increasing importance of financial markets, motives, institutions, and elites in the operation of the economy and its governing institutions. As a result, Chinese firms are becoming financialized, accentuating contact with global professional service corporations and “best practices” in human resource management and corporate governance. Moreover, human resource management now involves the incorporation of AI devices tracking productivity rates and producing warnings to employees concerning productivity delays.¹⁵⁰

Online lending and the rise of digital credit scoring are enabling shifts in China’s model of growth and development, thus empowering the development of new digital economies and markets which means that the role of finance and the banking system in China’s economic reform and development has played a central role. Additionally, the rise of DFS as a foundation for inclusive financial liberalization denotes a noteworthy move in the socio-economic reach of the Chinese financial system, along with newly shaped financial networks and relationships.

In other words, there is a move from the traditional network of commercial banks to those overseeing and exploiting the infrastructure of digital finance itself, and so, databases and data-processing techniques are embraced. Hence, the rise of Chinese DFS establishes a digital financialization process embedding technological innovation within the distinctive political economy of China and exploiting big data technologies, an economic growth model necessitating a competent financial system and a retail market environment for digital financial services that maximizes new openings for socio-economic development while minimizing

interconnected threats to systemic stability. On the other hand, digital financialization is generating new instruments for the Chinese government to supervise both individual and corporate players and employing algorithmic systems.¹⁵¹

1.13 CBDCs and Global Finance

Private cryptocurrencies and the upcoming CBDC lead to greater currency competition, generating regulatory complexities. There is a global drive to advance the efficiency of cross-border payments. CBDCs are a digital form of fiat money issued by a central bank, and currency substitution diminishes monetary authorities' control over national liquidity by limiting the component over which the authorities have direct influence. Hence, substitution into CBDCs is no different from conventional "dollarization" taking place in countries that have experienced high inflation and large exchange rate volatility. Therefore, the convenience and easier accessibility of CBDCs facilitate substitution at a faster pace and larger scale. It could be said that the CBDC could augment the degree of currency substitution in states that already use a foreign currency, which means that CBDCs could permit superior control of capital flows, depending on how they are planned and the degree of cooperation between the issuer and recipient state.

The increasing approval of CBDCs stems from numerous factors,¹⁵² but their widespread implementation and prospective implications have introduced concerns regarding the future global financial system. Consequently, the present financial background is certifying an unprecedented divergence. As more CBDCs circulate in the global market, the increased cybersecurity risk and high monetary mobility would have an effect on financial stability. CBDCs generate cybersecurity concerns that vary from those in the current digital financial system covering centralised financial data, problems with regulatory agencies, augmented complexity in client key custody, dependence on the integrity of third-party validators along with difficulty in transaction cancellation, and the prospective for intensified errors in scope and scale as a result of programmable transactions.¹⁵³

Furthermore, CBDCs vest central banks with notable capability to control monetary policies as they will have abundant data about countrywide transactions, but such a data-collecting-and-analyzing practice would subject central banks to a noteworthy privacy invasion concern.¹⁵⁴ Hence, cross-border transactions will compel the regulatory necessities for CBDC security. Thus, the global AML/CFT requirements for CBDC will concentrate on the interoperability of different CBDC systems and the capability to trace money flows through financial intermediaries.

It has to be taken into account that the circulation of different CBDCs in the global financial network will necessitate central banks to prepare new rules. Cheng-Yun, Tsang, and Ping-Kuei Chen¹⁵⁵ argue that “Geopolitical concerns on currency sovereignty and competition over fintech innovations can simultaneously encourage cooperation and confrontation. The central banks, financial intermediaries, and private sector should be ready to cope with significant changes in the global financial order.”

It has to be taken into account that the nature and pace of technological development are altering the face of globalization and underlining that trade in nonmaterial goods is subject to rapid technological innovation via distributed ledger technology. Hence, the digitization of trade in goods and services encompasses the implementation of widespread automated contracts, and so, the future launch of CBDCs for such cross-border payment transactions further stimulates the utilization of automated contracts.

Digitalization simplifies cross-border use of currencies, restructuring the demand for and supply of safe assets. Moreover, in terms of demand, an unequal rate of technological innovations across states or currency blocs, the appearance of alternative cross-border payment “rails,” or a transition to financial intermediation designated in a CBDC or global stablecoin could shift reserve currencies. In addition, in terms of supply, new digital platforms could emerge and reach a global scale, presenting alternative networks that CBDCs may tap into with the purpose of stimulating adoption upon issuance. Furthermore, CBDCs boost domestic payment systems by generating security and transparency within the monetary system, reducing the risks of fraud and money laundering, which means that augmenting

transparency in the flow of money contributes to lowering currency substitution.

It has to be taken into account that in the area of global finance, technological advancement alters the interacting dynamics among states and, as a result, the appearance of fintech has boosted certain countries to the forefront of global financial centers. However, presently the competition among fintech centers has not altered the global financial order. Concerning technology functioning within a jurisdiction, Przemysław Pałka¹⁵⁶ argues that “there is always a political decision taken. Regardless of whether it occurs explicitly through legislation, or implicitly, by leaving the decision to the market, and allowing corporations to write their own rules, some decision occurs.” On the other hand, CBDCs are deeply interlinked with global monetary territory and circulation, having the prospective to redirect global capital flows when they become prevalent, not mentioning the administration of banking by AAI technology functioning entirely on objectively technical terms without any political intervention will generate new financial centers grounded on the level of AAI technology utilized in various parts of the globe.

The rise of CBDCs is driven by technological developments in addition to geo-financial concerns.¹⁵⁷ Moreover, technological evolution and geo-financial inferences supplemented extra complexity to the present global financial order, restructuring the order. To that extent, countries are at different phases of financial market development, and so, the interacting dynamics between cryptocurrency and fiat currency remain unresolved by many sovereigns. In line with this, Cheng-Yun, Tsang, and Ping-Kuei Chen¹⁵⁸ argue that

“States might utilize CBDCs to fine-tune their monetary policy or even enable new forms of surveillance capitalism. To present a balanced view, we also highlighted that harmonization could be driven by factors such as central banks’ emphasis on the importance of smooth cross border transactions, the enforcement of AML/CFT rules, and the desire to achieve a more stable financial system...CBDC is

much more than a means of exchanging commodities and services. Developing CBDCs is hardly a domestic matter, especially for major global economies. Once a major economy launches CBDC and circulates globally, it will soon imply global financial governance.”

It has to be taken into consideration that the CBDC will compete with bank deposits in the market for digital money, which means that the CBDC will crowd out bank deposits, undermining financial stability by lifting the funding cost and diminishing the profitability of the banking sector. Hence, the CBDC escalates the deposit demand elasticity with regard to the deposit rate because households now have more options.¹⁵⁹

Taking into account that CBDC will have official government money status, CBDC will become a critical alternative to cash or bank credit. CBDC amplifies deposit spreads in this case, making more depositors choose shadow banks rather than commercial banks. Moreover, loan origination moves toward shadow banks, which are gaining share in a reducing market; thus, transferring loans to shadow banks will cause a critical allocation of credit risk in a rather vulnerable and unregulated part of the financial system. To that extent, Dmitry Oskolkov¹⁶⁰ argues that

“CBDC is shown to neutralize shadow monetary transmission channel when banks are unconstrained. While deposit spread increases in government bond interest rate, it declines in CBDC rate...increase in CBDC interest rate can act in the opposite manner to increase in government bond rate. This indicates that CBDC rate can become a useful instrument for monetary policy conduction...Firms face increasing loan rates when CBDC rate grows, but pay less taxes when CBDC rate is not big.”

Owing to its digital and centralized nature, CBDC is programmable, and those who govern the infrastructure can alter its qualities, which means that a central bank can adjust the interest rates, holding limits, and employing

restrictions. Moreover, due to their quality, both data and financial assets flow around the globe and have a cross-border character. Also, the success of CBDC for cross-border transactions depends on the access and interoperability selections made by central banks; therefore, technological and legal interoperability is central to CBDC's success, which means that a workable legal framework is needed for the establishment of a stable financial market infrastructure.

CBDC networks generate increased currency competition and so, CBDC networks possibly instigate the risk of increased uncertainties.¹⁶¹ On the other hand, the decentralized structures make it hard to instigate strong governance measures to stabilize the interchange mechanisms and the digital identity frameworks. Nonetheless, CBDC networks advance financial inclusion.¹⁶² Hence, CBDC will augment central banks' effectiveness in conducting monetary policy operations.¹⁶³

It is worth noting that access to cash remains vital to the public since it is their only alternative to commercial bank money, an alternative that becomes vital in times of distress for the banking sector. To that extent, CBDC will permit central banks to promote further reductions in cash usage.¹⁶⁴

As discussed earlier, E-money refers to an "electronic store" of monetary value on a technical device utilized for making payments.¹⁶⁵ Transformations in money have been driven by new technology and preferences, economic growth, and the need to fulfill socioeconomic change. Moreover, digital money, unlike e-money, has no parallel physical embodiment functioning in a technological environment. A digital and particularly a tokenized system presents advantages over an electronic system, and so, payments can be unified, reducing the costs linked with intermediaries.

It is characteristic that increased payment transparency diminishes money laundering and tax evasion. It must be taken into account that a common concern about CBDC is that transparency lessens tax evasion and money laundering, but, on the other hand, it presents risks to privacy and the misuse of the extraordinary new transactional data being generated and recorded in almost immediate reconciled time. Thus, digital currencies entail extremely sound and transparent economic governance by central

banks, and very transparent systems for controlling and retrieving the resulting new economic data.¹⁶⁶

Retail CBDCs are seen as digital versions of fiat money driven by the need to maintain the effectiveness of monetary policy, while the use of cash declines and interest in private money flourishes, there is a desire of advancing transaction efficiency and so improving accessibility and financial inclusion.¹⁶⁷ Also, CBDCs function as both a medium of exchange and a store of value, coexisting with other forms of central bank money and with commercial bank and e-money which means that CBDCs will serve as a up-to-date counterpart to physical cash.¹⁶⁸ It seems that CBDCs systems' design is the capacity to conduct transactions even without network connectivity¹⁶⁹ and so, Offline CBDC transactions will ensure access to payment functionalities in the absence of a reliable network connection or for the period of broader system failures.¹⁷⁰ It is impractical to preserve anonymity if a CBDC is issued, and the public nature of a CBDC would place it at the top of the financial pyramid. In this context, P. Michalopoulos et al.¹⁷¹ argue that “offline CBDC transactions with existing hardware and software technology solutions introduce additional degrees of flexibility to privacy-related designs that can even emulate the privacy features of physical cash.”

1.14 ChatGPT and Finance

ChatGPT is transforming traditional processes while causing unique challenges, and in finance, ChatGPT restructures customer interactions by advancing personalized financial advice, supporting investment strategies, and accelerating real-time market analysis. Hence, ChatGPT processes huge data speedily, boosting algorithmic trading, risk management, and fraud detection. Also, in accounting, AI systems automate data entry, categorization, and report generation, reducing human errors and operational costs and paving the road for supporting forensic accounting techniques. Nonetheless, ChatGPT and AI in finance and accounting are posing ethical dilemmas regarding data privacy, security, and biased decision-making algorithms, which means that there is a need to make certain that AI systems adhere to industry standards and regulations while

preserving the integrity and confidentiality of sensitive financial data. Data security is crucial, and strong cybersecurity measures protect sensitive financial data. In other words, in finance, ChatGPT can present personalized financial advice and support investment strategies by enabling real-time market analysis. Additionally, ChatGPT can handle big data promptly, manage risk, detect fraud, and enhance algorithmic trading. Rane, Nitin¹⁷² argues that

“While ChatGPT can analyze data and offer insights, final decisions in finance often carry profound implications. Striking the right balance between human judgment and AI-driven recommendations is vital to avoid excessive reliance on automated systems, ensuring that human expertise remains central in critical financial decisions.”

It has to be taken into account that there are concerns about the accountability of AI-driven financial decisions, necessitating a balance between human expertise and machine intelligence. Furthermore, the constant advancement of financial markets requires adaptability and continuous learning from AI systems.

ChatGPT has brought effectiveness, accuracy, and innovation, transforming the financial industry by processing vast datasets, automating tasks, and presenting valuable insights. Processes such as data entry, reconciliation, and financial analysis are streamlined, which means empowering finance authorities to concentrate on strategic decision-making.¹⁷³ To that extent, AI-powered chatbots boost customer service, guaranteeing real-time query resolution.

It is worth noting that financial reporting encompassing budgeting and cost analysis follows accounting standards, gains from AI automation, certifying accuracy and compliance. Moreover, there is a demand for transparency in AI algorithms guaranteeing that decisions made by AI systems are explainable and unbiased. Also, AI assists strategic planning via systematic analysis of financial records and so maintaining financial integrity. In line with this, Rane, Nitin¹⁷⁴ argues that “ChatGPT can indeed

catalyze positive change, propelling the finance and accounting industries toward a future marked by efficiency, accuracy, and ethical excellence.”

In addition, taxation gains from AI computation ensure compliance with changing laws. Moreover, regulatory compliance necessitates aligning AI with many financial regulations.

1.15 Blockchain and AI in Finance

It has to be taken into consideration that the convergence of blockchain technology and AI is steering in a new age of security and transparency within the financial sector, which means that this mixture forwards critical encounters faced by conventional financial systems while also displaying inventive solutions to enhance efficiency, minimize fraud, and strengthen transparency. Blockchain has to make certain that once data is recorded, it remains unaffected by modification, thus delivering an unparalleled level of security that seems to be challenged by new technologies. Therefore, blockchain must become an ideal choice for reinforcing financial systems against cyber threats and fraudulent activities. On the other hand, AI contributes predictive analytics and automation to the forefront of financial operations, enabling real-time data analysis, risk assessment, and decision-making, heightening processes and uplifting overall effectiveness. Thus, blockchain in tandem with AI should boost the precision and dependability of financial data, fostering a more secure and transparent ecosystem. To that extent, Rane, Nitin et al.¹⁷⁵ argue that

“the transparency ushered in by Blockchain ensures that all stakeholders have access to a singular version of the truth, fostering trust in financial transactions. Furthermore, the incorporation of AI in fraud detection and risk management heightens the proactive identification of potential threats, safeguarding financial institutions and their clientele.

It is characteristic that GNN-based fraud detection models must adhere to AML, KYC, and GDPR regulations, which means that fraud detection systems must be transparent and auditable.

In other words, the merging of blockchain and AI technologies is triggering a substantial modification in the financial landscape, and so, financial transactions and data move globally quickly, demanding a secure and transparent financial ecosystem. In line with this, Rane, Nitin et al.¹⁷⁶ say that

“Beyond security and transparency, the integration of Blockchain and AI redefines the nature of financial interactions. Smart contracts, powered by Blockchain, automate and enforce agreements without intermediaries, expediting processes and reducing the risk of disputes. AI-driven robo-advisors and personalized financial services leverage predictive analytics to offer tailored recommendations, enhancing customer experiences and financial outcomes.”

1.16 EU AI Act and Finance

The EU AI Act advances the embracing of AI systems in the EU by designing a legal framework that offsets the economic opportunities generated by AI with the demand to tackle the risks that the use of AI produces in terms of protecting public values¹⁷⁷ and the rule of law.¹⁷⁸ Moreover, the act relies on an ex-ante assessment by the lawmaker of the risks linked with some AI technologies. Additionally, Article 10(5) establishes a new legal basis for processing sensitive personal data on the basis that existing rules under EU data protection law were not enough to permit measures to diminish prospective biases in data utilized to train algorithms in high-risk contexts. Thus, the EU AI Act reveals a range of compromises and trade-offs.¹⁷⁹

It has to be taken into account that the EU AI Act applies horizontally and its horizontality means all domains covered by EU law, and so, AI systems used by financial institutions¹⁸⁰ and by the financial sector are covered by the Act which means that systems used to evaluate creditworthiness, credit scores, and risk assessments fall into the high-risk

category. Hence, the EU AI Act integrates its new obligations into standing procedures for financial supervision.¹⁸¹ In addition, the EU AI Act promotes the use of AI while preserving public values and forcing national authorities to adopt measures to support innovation. To that extent, Marco Almada¹⁸² argues that

“In order to carry out their duties in the age of AI, financial supervisory authorities need to develop additional capabilities. They need to foster technical competencies in AI, and the legal expertise required to navigate risks to fundamental rights that fall outside their traditional remit.”

It is characteristic that the EU AI Act¹⁸³ initiates a new category of AI systems named GPAI models, exposing considerable generality, but they accomplish a wide range of missions by prioritizing transparency and accountability and so, presenting technical documentation, instructions for use, and revealing training data while respecting copyright, which means lessening prospective risks and advancing trust within the AI society.¹⁸⁴ Furthermore, the EU AI Act states that institutions are subject to governance prerequisites and processes under EU financial law, thus accomplishing certain obligations defined in the regulation, subject to particular provisions regarding risk management and market monitoring. In fact, financial institutions have to keep documentation under EU law on financial services.¹⁸⁵ Besides, Kilian, Robert¹⁸⁶ et al. argue that

“the EU risks creating an uneven playing field that could significantly hinder innovation in the European AI ecosystem. Additionally, the amount and complexity of technical standards considered or developed as standardization deliverables should be reduced to make them more practicable.”

Furthermore, the EU AI Act endorses regulatory frameworks to keep pace with technological advancements, and the risks linked with GPAI systems

extend far beyond the narrow alarms of data quality and bias encompassing copyright violations, the misuse of AI-generated content, the creation of deepfakes, and the range of misinformation and disinformation utilized in creating highly realistic fake images and videos, setting contests for regulators due to the difficulties in distinguishing between human and synthetic content, particularly when it is distributed at scale.

Notes

1. X. Zhu, S. Asimakopoulos, and J. Kim, Financial Development and Innovation-led Growth: Is Too Much Finance Better? 100 *Journal of International Money and Finance* 102083 (2020). ↩
2. D. Kimani, K. Adams, R. Attah-Baakye, S. Ullah, J. Frecknall-Hughes, and J. Kim, Blockchain, Business and the Fourth Industrial Revolution: Whence, Whither, Wherefore and How?, 161 *Technological Forecasting and Social Change* 120254 (2020). ↩
3. Jessica E. Lees, Regulatory Influence in the Financial Markets Revisited, <https://ssrn.com/abstract=4804768>. ↩
4. Olga T. Cherney, Zhanna V. Smirnova Alexander P. Garin, Elena V. Romanovskaya, and Yuri A. Sidorenko, The Main Components of the Innovation Audit of the Economy, in E. N. Makarenko et al. (eds), *Technological Trends in the AI Economy, Smart Innovation, Systems and Technologies* (Singapore: Springer Nature Pvt Ltd., 2023), https://doi.org/10.1007/978-981-19-7411-3_12 P117. ↩
5. Saule T. Omarova, New Tech v. New Deal: Fintech as a Systemic Phenomenon, 36 *YALE J. ON REGUL.* 735 (2019). ↩
6. Saule T. Omarova, Technology v Technocracy: Fintech as a Regulatory Challenge, 6 *J. FIN. REGUL.* 75, 76 (2020). Saule T. Omarova, The People's Ledger: How to Democratize

- Money and Finance the Economy, 74 *Vanderbilt Law Review* 1232, <https://scholarship.law.vanderbilt.edu/vlr/vol74/iss5/1>.[↵]
7. Kenneth Krushel, Geopolitics in the Era of AI: Upending assumptions, ESCP Impact Paper N°2022-13-EN P101.[↵]
 8. N. Economides, The Economics of Networks, 14(6) *International Journal of Industrial Organization* 673–699 (1996).[↵]
 9. Satoshi Nakamoto, Bitcoin: A Peer-To-Peer Electronic Cash System 1 (2016), <https://bitcoin.org/bitcoin.pdf>
<https://perma.cc/H96W-5LJ6> Proof of Work, BITCOIN WIKI, https://en.bitcoin.it/wiki/Proof_of_work.[↵]
 10. Leah McGrath Goodman, The Face Behind Bitcoin, *NEWSWEEK* (Mar. 6, 2014), <http://www.newsweek.com/2014/03/14/face-behind-bitcoin-247957.html>. L. S., How Bitcoin Mining Works, *The Economist* (Jan. 20, 2015), <http://www.economist.com/blogs/economistexplains/2015/01/economist-explains-11>.[↵]
 11. Jonathan M. Barnett, The Host’s Dilemma: Strategic Forfeiture in Platform Markets for Informational Goods, 124 *Harv. L. Rev.* 1861, 1907 (2011) (“Like other successful open source projects, Linux code development is governed by a strict hierarchy, in which a limited core of qualified developers... develop code and approve changes to the code. These core developers are in turn assisted by reports of ‘bugs’ and ‘fixes’ contributed by a larger mass of participants.”).[↵]
 12. ECLI:ES:TS:2019:2109, <http://www.poderjudicial.es/search/indexAN.jsp>.[↵]
 13. L. Van Hove, Metcalfe’s Law and Network Quality: An Extension of Zhang et al., 37(1) *Journal of Computer Science and Technology* 117–123 (January 2016).[↵]

14. Case C-264/14, Hedqvist case, 2015, ECLI:EU:C:2015:718, fot. 12.↵
15. CJEU C-389/17, Case Paysera, 2019, ECLI:EU:C:2019:25, fot. 24.↵
16. Opinion of Advocate General M. WATHELET, 2018, Case C-389/17, Paysera LT judgment, fot.53.↵
17. John O. McGinnis, Two Paradoxes of Crypto, 2023 *Chapman Law Review* Vol. 26:2 /445 p450, 451.↵
18. R. April, A Proposal of a New Pattern for a Land Registry Based on Blockchain Technology, 28 *Derecom* 65–94 (2020).↵
19. M. Scharnowski, S. Scharnowski, and L. Zimmermann, Fan Tokens: Sports and Speculation on the Blockchain, 89 *Journal of International Financial Markets, Institutions and Money* (2023), <https://doi.org/10.1016/j.intfin.2023.101880>.↵
20. Betsy Vereckey, Who Will Regulate Crypto and FinTech in the US?, *Mit Mgmt. Sloan School* (Nov. 8, 2022), <http://perma.cc/Q5CP-ENF2>.↵
21. Michael Conklin, Driving a Stake Through Corporate Cryptocurrency Involvement: Why the Practice Fails the Stakeholder Ethical Analysis, 69 *Wayne Law Review* 145, P146, 168 (2024).↵
22. D. A. Wallach, Bitcoin for Rockstars: How Cryptocurrency Can Revolutionize the Music Industry, *MEDIUM* (Dec. 10, 2014), <https://medium.com/backchannel/bitcoin-for-rockstarsca8366802f9>.↵
23. BITWAGE, www.bitwage.co (a Bitcoin-based payment system allowing employees to be paid in micro amounts over a pay period).↵
24. Crypto-Currency Market Capitalizations, <https://coinmarketcap.com/>.↵

25. *Contracts*, Bitcoin Foundation WIKI,
<https://en.bitcoin.it/wiki/Contracts>.[↵]
26. *Market Capitalization*, Blockchain,
[https://blockchain.info/charts/market-cap?
timespan=all&daysAverageString=1&scale=1&address=](https://blockchain.info/charts/market-cap?timespan=all&daysAverageString=1&scale=1&address=).[↵]
27. Stephen T. Middlebrook and Sarah Jane Hughes, Regulating Cryptocurrencies in the United States: Current Issues and Future Directions, 40 *Wm. Mitchell L. Rev.* 813, 819–820 (2014).[↵]
28. Sarah Murray and Elizabeth Butler, Cryptocurrency—Is It Recognised as “Property” under English Law?, *Stevens & Bolton* (Jun. 10, 2022).[↵]
29. English High Court in *AA v. Persons Unknown and Others*, Re Bitcoin [2019] EWHC 3556 (Comm).[↵]
30. [2020] SGCA(I) 02.[↵]
31. *Ruscoe v Cryptopia Ltd (in Liquidation)* [2020] NZHC 728.[↵]
32. Adetomiwa Fowowe, Cryptocurrency: Legal Analysis on the Future of Financing International Commerce,
<https://ssrn.com/abstract=4792357> p. 14. A. I. Fowowe, The Governance or Otherwise of Cryptocurrency in the Sales of Goods Act, 1893, (2021), <https://ssrn.com/abstract=3833672>.[↵]
33. New York Times, Whispers of Return to Drachma Grow Louder in Greek Crisis. By Landon, T, Nov. 1st Edition (2011).[↵]
34. Paul Krugman, Bitcoin Is Evil, *New York Times*, December 28th Edition (2013).[↵]
35. N. R. Kocherlakota, Money Is Memory, 81 *Journal of Economic Theory* 232–251 (1998).[↵]
36. Maria Bigoni, Gabriele Camera, and Marco Casari, Money Is More Than Memory, October 27, 2014 No. 496, P 31–33.[↵]
37. G. Zekos, Monetary Policy and Risk Management in Financial Globalization (New York: Nova Science Publications, 2015),

<https://novapublishers.com/shop/monetary-policy-and-risk-management-in-financial-globalization/>.[↵]

38. Adam J. Levitin, *Safe Banking*, 2015, <http://ssrn.com/abstract=2532703>.[↵]
39. Morgan Ricks, *Entry Restriction, Shadow Banking, and the Structure of Monetary Institutions*, 2 *J. FIN. REG.* 291 (2016).[↵]
40. Morgan Ricks, *Money, Private Law, and Macroeconomic Disasters*, 83 *Law and Contemporary Problems* P81 (2020).[↵]
41. Hans Gersbach, Hans Haller, and Stylianos Papageorgiou, *Regulatory Competition in Banking: Curse or Blessing?* 2020 *Journal of Banking & Finance* 105954 (2020).[↵]
42. Article 2.e of Directive 2009/110/EC of the European Parliament and of the Council of 16 September 2009 on the taking up, pursuit, and prudential supervision of the business of electronic money institutions, amending Directives 2005/60/EC and 2006/48/EC and repealing Directive 2000/46/EC.[↵]
43. European Cent. Bank, *Virtual Currency Schemes—A Further Analysis* 15 (2015), <https://www.ecb.europa.eu/pub/pdf/other/virtualcurrencyschemes.pdf>.[↵]
44. 31 C.F.R. § 1010.100(m) (2016). U.S. Dep’t of the Treasury, *Financial Crimes Enforcement Network, Guidance Note FIN-2013-G001* (Mar. 18, 2013), <https://www.fincen.gov/sites/default/files/shared/FIN-2013-G001.pdf>.[↵]
45. U.S. Dep’t of the Treasury, *Financial Crimes Enforcement Network, Guidance Note FIN-2013-G001* (Mar. 18, 2013), <https://www.fincen.gov/sites/default/files/shared/FIN-2013-G001.pdf>. R. J. Straus, *The FinCEN Virtual Currency*

Guidance: Neutering Bitcoin?, 7(4) *E-Finance & Payments Law & Policy* 9 (2013).⁴⁶

46. SEC Chairman Jay Clayton and CFTC Chairman J. Christopher Giancarlo, Regulators Are Looking at Cryptocurrency: At the SEC and CFTC We Take Our Responsibility Seriously, *Wall Street Journal* (Jan. 24, 2018), <https://www.wsj.com/articles/regulators-arelooking-at-cryptocurrency-1516836363>.⁴⁷
47. Securities and Exchange Commission, Office of Investor Education and Advocacy, Ponzi Schemes Using Virtual Currencies, July 2013. Securities and Exchange Commission, Office of Investor Education and Advocacy, Investor Alert: Bitcoin and Other Virtual Currency-Related Investments (May 7, 2015), <https://investor.gov/additional-resources/news-alerts/alerts-bulletins/investor-alert-bitcoin-other-virtual-currency>.⁴⁸
48. Joint Statement by SEC and CFTC Enforcement Directors Regarding Virtual Currency Enforcement Actions, Jan. 19, 2018. SEC Press Rel. 2018-23, Feb. 21, 2018. SEC Files Fraud Charges in Bitcoin and Office Space Investment Schemes, SEC Rel. 2017-123, June 30, 2017.⁴⁹
49. A. Hayes, Competing Imaginaries: Crypto-utopianism and the Material Forces of Bitcoin Mining, 39(4) *Anthropol. Today* 4–8 (2023), <https://doi.org/10.1111/1467-8322.12825>.⁵⁰
50. SEC v. Shavers, 2014 WL 4652121 (E.D. Tex. 2014);⁵¹
51. *In re Coinflip, Inc.*, CFTC No. 15-29 (Sept. 17, 2015) (finding that digital currency qualifies as a commodity under 7 U.S.C. § 1a(9)).⁵²
52. *In re TeraExchange LLC*, CFTC Docket No. 15-33 (Sept. 24, 2015);⁵³
53. *Investor Alert: Bitcoin: More Than a Bit Risky*, FINRA (May 7, 2014), <http://www.finra.org/investors/alerts/bitcoin-more->

bit-risky.↵

54. Security and Exchange Commission, Release No. 71568 (Feb. 19, 2014), <https://www.sec.gov/litigation/suspensions/2014/34-71568.pdf>.↵
55. SEC v. Shavers, No. 4:13-CV-416, 2013 WL 4028182, (E.D. Tex. 2013), *reconsideration aff'd*, No. 4:13-CV-416, 2014 WL 12622292 (E.D. Tex.2014).↵
56. 15 U.S.C. §§ 77t, 77v (2012). Registration Under the Securities Act of 1933, U.S. Sec. & Exch. Comm'n, <http://www.sec.gov/answers/regis33.htm> 15 U.S.C. §§ 78u, 78aa (2012).↵
57. SEC v. Garza, No. 3:15-CV-01760, 2015 WL 7732649 (D. Conn., Complaint filed Dec. 1, 2015) (“[Defendant’s Bitcoin products] constitute investment contracts and thus ‘securities’ under Section 2(a)(1) [15 U.S.C. §77b(1)] of the Securities Act.”).↵
58. SEC v. W.J. Howey Co., 328 U.S. 293 (1946). United States v. Faiella, 39 F. Supp. 3d 544 (S.D.N.Y. 2014); United States v. Budovsky, No. 13CR368 DLC, 2015 WL 5602853, at *1 (S.D.N.Y. 2015); United States v. Murgio, No. 15-CR-769 (AJN), 2016 WL 5107128 (S.D.N.Y. 2016) (none arguing that Bitcoin does not satisfy the “investment contract” definition).↵
59. 15 U.S.C. § 77a (1933). 15 U.S.C. § 78j(b) (2012). 17 C.F.R. § 240.10b-5(b) (2015). SEC v. Morgan Keegan & Co., Inc., 678 F.3d 1233, 1244 (11th Cir. 2012).↵
60. Dep’t of Justice, <https://www.justice.gov/opa/pr/odebrecht-and-braskem-plead-guilty-and-agreepay-least-35-billion-global-penalties-resolve>.↵
61. See Linda Chatman Thomsen, Deputy Director, Division of Enforcement, SEC, An Overview of Enforcement, SEC,

https://www.sec.gov/about/offices/oia/oia_enforce/overviewenfor.pdf.[↵]

62. Nathaniel Popper, S.E.C. Rejects Winklevoss Brothers' Bid to Create Bitcoin E.T.F., *THE N.Y. Times* (Mar. 10, 2017), https://www.nytimes.com/2017/03/10/business/dealbook/winklevoss-brothersbid-to-create-a-bitcoin-etf-is-rejected.html?_r=0.[↵]
63. Andrew Verstein, Crypto Assets and Insider Trading Law's Domain, 105 *Iowa L. Rev.* 1, 9 (2019); J. S. Nelson, Cryptocommunity Currencies, 105 *Cornell L. Rev.* 909, 934–935 (2020); Eric D. Chason, Smart Contracts and the Limits of Computerized Commerce, 99 *Neb. L. Rev.* 330, 363–364 (2020).[↵]
64. Robert Sistros, Post-Etherdelta: Clarifying Liabilities for Cryptocurrency Exchanges and Market Participants, 39 *Cardozo Arts & Ent. L.J.* 343, 354 (2021) (“If a cryptocurrency token is decentralized such that purchasers’ expectation of profits rests, not on a person or group, but in an autonomous network or structure, the token fails to satisfy the Howey test as an investment contract and would not constitute a security.”); *In re Tether & Bitfinex Crypto Asset Litig.*, 576 F. Supp. 3d 55, 71 (S.D.N.Y. 2021) (“USDT...[is] a ‘stablecoin,’ so called because it is purportedly pegged to and backed by U.S. dollars held in reserve by Tether.”).[↵]
65. Usha R. Rodrigues, Embrace the SEC, 61 *Wash. U. J.L. & Pol’y* 133, 147–148 (2020) (“For their part, ICO enthusiasts need to understand the SEC’s perspective. For decades, the SEC has labored to protect the general public from the hype of risky offerings, worried that grandma will lose her savings to unscrupulous promoters. The SEC has targeted extremely troubling offerings.”).[↵]

66. Stephen Morris and Hyun Song Shin, Financial Regulation in a System Context (2008), https://www.brookings.edu/wp-content/uploads/2008/09/2008b_bpea_morris.pdf.[↵]
67. Michael Woodford, Central Bank Communication and Policy Effectiveness, in *Proceedings of the Federal Reserve Bank of Kansas City Economic Symposium at Jackson Hole* (2005), <http://www.kc.frb.org/publicat/sympos/2005/sym05prg.htm>.[↵]
68. Milton Friedman and Anna J. Schwartz, Has Government Any Role in Money? <http://www.nber.org/chapters/c7507> Chapter pages in book: (p. 289—314) University of Chicago Press (1987) p. 311.[↵]
69. Frederic S. Mishkin, Monetary Policy Strategy: Lessons from the Crisis January 2011 ECB, p. 50, Central Banking Conference, Monetary Policy Revisited: Lessons from the Crisis, Frankfurt, November 18–19, 2010.[↵]
70. Yener Altunbas, Leonardo Gambacorta, and David Marqués-Ibáñez, Bank Risk and Monetary Policy Working Papers Number 712—May 2009.[↵]
71. Yener Altunbas, Leonardo Gambacorta, and David Marqués-Ibáñez, Bank Risk and Monetary Policy Working Papers Number 712—May 2009.[↵]
72. C. Borio and H. Zhu, Capital Regulation, Risk-Taking and Monetary Policy: A Missing Link in the Transmission Mechanism?, BIS Working Papers, No. 268 (2008).[↵]
73. N. Rajan, Has Financial Development Made the World Riskier?, NBER Working Paper Series No. 11728 (2005).[↵]
74. Wolfgang Breuer and Andreas Knetsch, Recent Trends in the Digitalization of Finance and Accounting, 93 *Journal of Business Economics* 1451–1461 (2023), <https://doi.org/10.1007/s11573-023-01181-5> p 1454.[↵]
75. R. Nunes, Adoption of Cryptocurrency and Abandonment of Monetary Policy: An Analysis in the Light of Economic

- Theory, 28(11) *IOSR Journal of Humanities and Social Science* 30–36 (2023).[↵]
76. Salam Alshareef, The Gulf's Shifting Geoeconomy and China's Structural Power: From the Petrodollar to the Petroyuan? 27(2) *Competition and Change*, <https://doi.org/10.1177/10245294221095222>.
<https://journals.sagepub.com/home/CCH> (The effectiveness of the petrodollar regime has been significantly reduced as China has overtaken the United States as the largest destination for the recycling of oil receipts from the GCC through the trade channel.)[↵]
 77. G. T. Chin and K. P. Gallagher, Coordinated Credit Spaces: The Globalization of Chinese Development Finance, 50(1) *Development and Change* 254–274 (2019).[↵]
 78. Osama Wagdi, Akram Elnahrawy, and Atef Fathi, Petrodollar and De-dollarization: A Survey from OAPEC Countries, 449 *E3S Web of Conferences* 02008 (2023),
<https://doi.org/10.1051/e3sconf/202344902008> p1.[↵]
 79. Yayaj. Fanusie and Emilyjin, China's Digital Currency: Adding Financial Data to Digital Authoritarianism 1 (2021).[↵]
 80. Lan Cao, Dollar Trap and Digital Currency, 26(2) *Chapman Law Review* 503 P555 (2023).[↵]
 81. M. Z. Iqbal and A. G. Campbell, Metaverse as Tech for Good: Current Progress and Emerging Opportunities, 2(4) *Virtual Worlds* 326–342 (2023).[↵]
 82. D. Vidal-Tomás, The Illusion of the Metaverse and Meta-economy, 86 *International Review of Financial Analysis* 102560 (2023).[↵]
 83. Codice civile, RD 16 March 1942, n. 262.[↵]
 84. Art. 65 French Loi n. 486 regulated (bien incorporel)
 Rampone, F., Token, between crypto-assets and other digital elements, 10/04/2020

(<https://www.blockchain4innovation.it/esperti/token-tra-crypto-asset-e-altri-elementi-digitali/>)

85. K. B. Ooi, G. W. H. Tan, E. CH. X. Aw, T. H. Cham, R. Dwivedi, L. Hughes, K. A. Kar, X. M. Loh, E. Mogaji, I. Phau, and A. Sharma, Banking in the Metaverse: A New Frontier for Financial Institutions, 41(7) *International Journal of Bank Marketing* 1829–1846 (2023).
86. Deniz Güven, Step Inside Our Hong Kong Virtual Bank, *Standard Chartered* (May 29, 2019), <https://www.sc.com/en/navigate-the-future/hong-kong-virtual-bank/> [<https://perma.cc/S8KL9G65>].
87. John Callahan, Know Your Customer (KYC) Will Be a Great Thing When It Works, *Forbes* (July 10, 2018), <https://perma.cc/L48R-PMFL>.
88. Combating Financial Crime, *KPMG* (Mar. 2019), <https://home.kpmg/xx/en/home/insights/2019/03/combating-financial-crime-fs.html> [<https://perma.cc/W99N-677A>].
89. M. MacDonald, The Intersection between IP and the Metaverse: Preliminary Observations, Queen Mary Law Research Paper No. 397/2023, January 26, 2023, <https://ssrn.com/abstract=4338884>. C. Macedo et al., The Metaverse: From Science Fiction to Commercial Reality—Protecting Intellectual Property in the Virtual Landscape, 31(1) *NYSBA* 13–20 (2022).
90. Gaetano Dimita, Yin Harn Lee, Michaela MacDonald, Anthony Michael Catton, Zeynep Kubra Kavcar Penbegullu, and Juan Alberto Pulido Lock, IP and Metaverse(s)—An Externally Commissioned Research Report, Queen Mary Law Research Paper No. 427/2024.
91. F. Mostert and W. Yeoh, Meta-worse, a Lawyer’s Mega Paradise, 17(3) *Journal of Intellectual Property Law &*

- Practice* 211–212 (2022),
<https://doi.org/10.1093/jiplp/jpac00>.[↵]
92. Andrea M. Matawyshn, Material Vulnerabilities: Data Privacy, Corporate Information Security, and Securities Regulation, 3 *Berkeley Bus. L.J.* 129, 145 (2005).[↵]
93. T. Huynh-The, Q. Pham, X. Pham, T. Nguyen, Z. Han, and D. Kim, Artificial Intelligence for the Metaverse: A Survey, 117 *Engineering Applications of Artificial Intelligence* 105581 (2023).[↵]
94. B. Hammi, S. Zeadally, and A. Perez, Non-fungible Tokens: A Review, 6(1) *IEEE Internet of Things Magazine* 46–50 (2023).[↵]
95. Chen Jun and Dr. Danny Friedmann, Jumping from Mother Monkey to Bored Ape: The Value of NFTs from an Artist’s and Intellectual Property Perspective,
<https://ssrn.com/abstract=4213536> P45.[↵]
96. Joerg Osterrieder, Stephen Chan, Yuanyuan Zhang, and Jeffrey Chu, Metaverse Non-Fungible Tokens, at:
<https://ssrn.com/abstract=4733153> p. 26[↵]
97. I. Yousaf, A. Abrar, and L. Yarovaya, Decentralized and Centralized Exchanges: Which Digital Tokens Pose a Greater Contagion Risk? *Journal of International Financial Markets, Institutions and Money* (2023),
<https://doi.org/10.1016/j.intfin.2023.101881>.[↵]
98. Suliman Alshahmy, Arnaz Binsardi, and Padmi Nagirikandalage, Relationship between the Pricing and Trading of Cryptocurrencies and the Collectible, Metaverse, and Art Non-Fungible Token Subcategories,
<https://ssrn.com/abstract=4801958> P6.[↵]
99. Weiwei Guo et al., *The British Accounting Review*,
<https://doi.org/10.1016/j.bar.2025.101641> p1.[↵]

100. R. Wilson Freyermuth, Christopher K. Odinet, and Andrea Tosato, Crypto in Real Estate Finance, 75(1) *Alabama Law Review* 93 P94 (2023).[↵](#)
101. Dirk A. Zetzsche, Douglas Arner, Ross Buckley, and Brian W. Tang, Artificial Intelligence in Finance: Putting the Human in the Loop, <https://ssrn.com/abstract=3531711> P3.[↵](#)
102. Margot E. Kaminski and Jennifer M. Urban, The Right to Contest AI, 121 *COLUM. L. REV.* 1957 (2021).[↵](#)
103. Brandon L. Garrett and Cynthia Rudin, The Right to a Glass Box: Rethinking the Use of Artificial Intelligence in Criminal Justice, 109 *Cornell Law Review* 561 p. 627 (2024).[↵](#)
104. Nico Lehnertz, Carolin Plagmann, and Eva Lutz, Why Deep Pockets Make Great Borrowers: An Empirical Analysis of Venture Loans, 92 *Journal of Business Economics* 1431–1453 P1431 (2022), <https://doi.org/10.1007/s11573-022-01084-x>.[↵](#)
105. Y. V. Hochberg, C. J. Serrano, and R. H. Ziedonis, Patent Collateral, Investor Commitment, and the Market for Venture Lending, 130 *J Financ Econ* 74–94 (2018), <https://doi.org/10.1016/j.jfneco.2018.06.003>[↵](#)
106. F. Bertoni, M. G. Colombo, A. Quas, and F. Tenca, The Changing Patterns of Venture Capital Investments in Europe, 46 *J Ind Bus Econ* 229–250 (2019), <https://doi.org/10.1007/s40812-019-00113-1>.[↵](#)
107. L. Zhang, Y. Guo, and G. Sun, How Patent Signals Affect Venture Capital: The Evidence of Bio-pharmaceutical Start-ups in China, 145 *Technol Forecast Soc Change* 93–104 (2019), <https://doi.org/10.1016/j.techfore.2019.05.013>.[↵](#)
108. Maximilian Germann and Christoph Merkle, Algorithm Aversion in Delegated Investing, 93 *Journal of Business Economics* 1691–1727 p169. 1 (2023), <https://doi.org/10.1007/s11573-022-01121-9>.[↵](#)

109. A. V. Thakor, Fintech and Banking: What Do We Know?, 41 *J Financ Intermed* 100, 833 (2020).[↵](#)
110. S. Raisch and S. Krakowski, Artificial Intelligence and Management: The Automation–augmentation Paradox, 46(1) *Academy of Management Review* 192–210 (2021).[↵](#)
111. S. M. Chew and F. Spiegl, Security Token Offering—New Way of Financing in the Digital Era, 52. *Journal of Financial Transformation* 142–151 (2021). D. A. Zetsche, D. W. Arner, and R. P. Buckley, Decentralized Finance, 6(2) *Journal of Financial Regulation* 172–203 (2020).[↵](#)
112. Yan Chen, Jack I. Richter, and Pankaj C. Patel, Decentralized Governance of Digital Platforms, 47(5) *Journal of Management* 1305–1337 p 1305 (May 2021).[↵](#)
113. T. Lambert, D. Liebau, and P. Roosenboom, Security Token Offerings, *Small Business Economics* (2021), <https://doi.org/10.1007/s11187-021-00539-9>.[↵](#)
114. L. Ante, Blockchain-based Tokens as Financing Instruments: Capital Market Access for SMEs?, in I. A. Boitan and K. Marchewka-Bartkowiak (eds), *Fostering Innovation and Competitiveness with FinTech, RegTech, and SupTech* (Hershey, PA: IGI Global, 2021), pp. 129–141.[↵](#)
115. R. Belchior, A. Vasconcelos, S. Guerreiro, and M. Correia, A Survey on Blockchain Interoperability: Past, Present, and Future Trends, arXiv preprint arXiv:2005.14282 (2020).[↵](#)
116. Wolfgang Breuer and Andreas Pfngsten, Non Standard Issues in Business Finance: An Overview, 92 *Journal of Business Economics* 1417–1430 (2022), <https://doi.org/10.1007/s11573-022-01122-8>.[↵](#)
117. A. Sunyaev, N. Kannengießner, R. Beck, H. Treiblmaier, M. Lacity, J. Kranz, G. Fridgen, U. Spankowski, and A. Luckow Token Economy, 63(4) *Business & Information Systems*

- Engineering* 457–478 (2021), <https://doi.org/10.1007/s12599-021-00684-1>.[↵]
118. Y. Chen, I. Pereira, and P. C. Patel, Decentralized Governance of Digital Platforms, 47(5) *Journal of Management* 1305–1337 (2020), <https://doi.org/10.1177/0149206320916755>.[↵]
 119. Y. Wang, Automated Market Makers for Decentralized Finance (DeFi), arXiv preprint arXiv:2009.01676 (2020).[↵]
 120. R. H. Weber, Non-fungible Tokens—A New (Legal) Phenomenon in the Crypto Universe. *Sic* 487–496.[↵]
 121. A. Whitaker and R. Kräussl, Fractional Equity, Blockchain, and the Future of Creative Work, 66(10) *Management Science* 4594–4611 (2020).[↵]
 122. World Economic Forum, & Accenture, Modernizing Financial Markets with Wholesale Central Bank Digital Currency (wCBDC) (2024), <https://www.weforum.org/publications/modernizing-financial-markets-with-wcbdc/>.[↵]
 123. Rosa M. Garcia-Teruel and Héctor Simón-Moreno, The Digital Tokenization of Property Rights. A Comparative Perspective, 41 *Computer Law & Security Review* 105543 (2021) p1, www.elsevier.com/locate/CLSR.[↵]
 124. Darian M. Ibrahim, A Tokenized Future: Regulatory Lessons from Crowdfunding and Standard Form Contracts, 74 *Hastings Law Journal* 45 p. 45 (2022).[↵]
 125. Barry Friedman, Farhang Heydari, Max Isaacs, and Katie Kinsey, Policing Police Tech: A Soft Law Solution, 37 *Berkeley Tech. L.J.* 701, 117 (2022). Niels van Dijk, Simone Casiraghi, and Serge Gutwirth, The ‘Ethification’ of ICT Governance. Artificial Intelligence and Data Protection in the European Union, 43 *Comput. L. & Sec. Rev.* 1, 7 (2021).[↵]
 126. D. Cumming, G. Leboeuf, and A. Schwienbacher, Crowdfunding Models: Keep-it All vs. All-or-nothing, 49(2)

- Financial Management* 331–360 (2020),
<https://doi.org/10.1111/fima.12262>.[↵]
127. G. Borello, V. De Crescenzo, and F. Pichler, Factors for Success in European Crowdfunding, 106 *Journal of Economics and Business* 1–19 (2019),
<https://doi.org/10.1016/j.jeconbus.2019.05.002>.[↵]
 128. S. Ribeiro-Navarrete, D. Palacios-Marqués, C. Lassala, and K. Ulrich, Key Factors of Information Management for Crowdfunding Investor Satisfaction, 59 *International Journal of Information Management* 1–10 (2021),
<https://doi.org/10.1016/j.ijinfomgt.2021.102354>.[↵]
 129. D. J. Cumming and L. Hornuf, *The Economics of Crowdfunding: Startups, Portals and Investor Behavior* (New York: Palgrave Macmillan, 2018).[↵]
 130. H. Tang, Peer-to-peer Lenders versus Banks: Substitutes or Complements? 32 *The Review of Financial Studies* 1900–1938 (2019).[↵]
 131. Z. Zhang, W. Hu, and T. Chang, Nonlinear Effects of P2P Lending on Bank Loans in a Panel Smooth Transition Regression Model, 59 *International Review of Economics & Finance* 468–473 (2019).[↵]
 132. W. Breuer, C. K. Soypak, and B. I. Steininger, Magnitude Effects in Lending and Borrowing: Empirical Evidence from a P2P Platform, *The European Journal of Finance* (2020),
<https://doi.org/10.1080/1351847X.2019.1709525>.[↵]
 133. Darian M. Ibrahim, Underwriting Crowdfunding, 25 *Stan. J.L. Bus. & Fin.* 289, 300–301 (2020) (“[S]tartups can only disclose so much given they are so young.”). Chris Brummer, Disclosure, Dapps and DeFi, 5 *Stan. J. Blockchain L. & Pol’y* 137, 147 (2022) (“[T]he document[s] initial issuers of securities file with the SEC to disclose key facts about their business[] fail to anticipate decentralized architectures, and are

both over- and under-inclusive in terms of the disclosure requirements that one would expect of issuers of blockchain-based securities.”).¹³⁴

134. Georgios Dimitropoulos, *The Law of Blockchain*, 95 *Wash. L. Rev.* 1117, 1128 (2020) (“Before a block can be added to the chain, a cryptographic puzzle must be solved, creating the block. Miners are special nodes that place transactions in a block by successfully solving a Proof of Work (PoW) or other problems.”).¹³⁵
135. Andrew A. Schwartz, *Mandatory Disclosure in Primary Markets*, 2019 *UTAH L. REV.* 1069, 1076 (2019) (“Mandatory disclosure imposes such significant costs that it deters many companies, especially small ones, from conducting an IPO in the first place.”).¹³⁶
136. Cumming, D., G. Leboeuf, and A. Schwienbacher, *Crowdfunding Models: Keep-It-All vs. All-Or-Nothing*, 49(2) *Financial Management* 331–360 (2020), <https://doi.org/10.1111/fima.12262>.¹³⁷
137. Jerry Coakley and Winifred Huang, *P2P Lending and Outside Entrepreneurial Finance*, 29(13) *The European Journal of Finance* 1520–1537 (2023), <https://doi.org/10.1080/1351847X.2020.184222> P1520.¹³⁸
138. Arun Adwani, *Foundations of Fintech: Navigating the Digital Financial Revolution* (Red Unicorn Publishing Pvt. Ltd., 2025).¹³⁹
139. Julian Franks, Nicolas Serrano-Velarde, and Oren Sussman, *Marketplace Lending, Information Aggregation, and Liquidity*, Finance Working Paper No. 678/2020 p 1.¹⁴⁰
140. Ressita Ramadhani, Zakka Farisy, and Dina Silvia Puteri, *Comparative Analysis of CBDC and Tax Law Enforcement in Selected Countries*, 4(1) *Journal of Central Banking Law and Institutions* 141–180 (2025).¹⁴¹

141. D. A. Zetsche, D. W. Arner, and R. P. Buckley, Decentralized Finance, 6(2) *Journal of Financial Regulation* 172–203 (2020).[↵](#)
142. D. G. Baur, K. Hong, and A. D. Lee, Bitcoin: Medium of Exchange or Speculative Assets? 54 *Journal of International Financial Markets, Institutions and Money* 177–189 (2018), <https://doi.org/10.1016/j.intfin.2017.12.004>.[↵](#)
143. V. I. Gromyko and N. S. Vasilyev, Mathematical Modeling of Deep-learned Artificial Intelligence and Axiomatic for System-informational Culture, 4(4) *Int. J. Robot. Autom.* 245–246 (2018).[↵](#)
144. A. McAfee, E. Brynjolfsson, T. H. Davenport, D. J. Patil, and D. Barton, Big Data: The Management Revolution, 90(10) *Harvard Bus. Rev.* 60–68 (2012). X. Wu, X. Zhu, G. Q. Wu, and W. Ding, Data Mining with Big Data, 26(1) *IEEE Trans. Knowl. Data Eng.* 97–107 (2014). A. A. Tole, Big Data Challenges, 4(3) *Database Syst. J.* 31–40 (2013).[↵](#)
145. U. S. Förster-Metz, K. Marquardt, N. Golowko, A. Kompalla, and C. Hell, Digital Transformation and its Implications on Organizational Behavior, *J. EU Res. Bus.* (2018), <https://doi.org/10.5171/2018.340873>.[↵](#)
146. V. Chung, M. Dietz, I. Rab, and Z. Townsend, McKinsey Quarterly Ecosystem 2.0: Climbing to the Next Level (2020), <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/ecosystem-2-point-0-climbing-to-the-next-level#>.[↵](#)
147. S. Claessens, J. Frost, G. Turner, and Z. Feng, Fintech Credit Markets Around the World: Size, Drivers and Policy Issues, *BIS Quarterly Review*, September: 29–49 (2018).[↵](#)
148. Julian Gruin, Financializing Authoritarian Capitalism: Chinese Fintech and the Institutional Foundations of Algorithmic Governance, 5(2) *Finance and Society* 84–104 P85 (2019).[↵](#)

149. J. Gruin, *Communists Constructing Capitalism: State, Market, and the Party in China's Financial Reform* (Manchester: Manchester University Press, 2019).↵
150. A.-S. Mayer, F. Strich, and M. Fiedler, Unintended Consequences of Introducing AI Systems for Decision Making, 19(4) *MIS Quarterly Executive* 237–257 (2020).↵
151. J. Wang, Inclusion or Expulsion: Digital Technologies and the New Power Relations in China's 'Internet Finance', 3(1) *Communication and the Public* 34–45 (2018).↵
152. Tao Zhang, New Forms of Digital Money: Implications for Monetary and Financial Stability (2020), <https://www.imf.org/en/News/Articles/2020/10/30/sp103020-new-forms-ofdigital-money>.↵
153. Andrew Stanley, The Ascent of CBDCs, IMF, September 2022, <https://www.imf.org/en/Publications/fandd/issues/2022/09/Picture-this-The-ascent-of-CBDCs>.↵
154. IMF, IMF Approach to Central Bank Digital Currency Capacity Development. Policy Papers 2023(016) (2023), <https://www.elibrary.imf.org/view/journals/007/2023/016/article-A001-en.xml>. Michael Lloyd and Chris Dixon, A Future Multipolar World, 13(5) *Global Policy* 818–827 (2022).↵
155. Cheng-Yun (CY) Tsang and Ping-Kuei Chen, Global Financial Order at a Crossroads Do CBDCs Lead to Balkanisation or Harmonization?, <https://ssrn.com/abstract=4636943> p1.↵
156. Przemysław Pałka, Data Management Law for the 2020s: The Lost Origins and the New Needs, 68 *Buff. L. Rev.* 559 (2020) P640, <https://digitalcommons.law.buffalo.edu/buffalolawreview/vol68/iss2/4>.↵
157. Brad W. Setser, The New Geopolitics of Global Finance, Council on Foreign Relations (2022), <https://www.cfr.org/blog/new-geopolitics-global-finance>.↵

158. Cheng-Yun (CY) Tsang and Ping-Kuei Chen, Global Financial Order at a Crossroads Do CBDCs Lead to Balkanisation or Harmonization?, <https://ssrn.com/abstract=4636943> P15–16.↵
159. Jiaqi Li, Andrew Usher, and Yu Zhu, Central Bank Digital Currency and Banking Choices, <https://ssrn.com/abstract=4705512>.↵
160. Dmitry Oskolkov, The Effect of CBDC on Competition Between Shadow and Commercial Banks, <https://ssrn.com/abstract=4708658> p16–17.↵
161. Dionysopoulos Lambis/Marra Miriam/Urquhart Andrew, Central Bank Digital Currencies: A Critical Review, February 2023.↵
162. Zhang Tao, New Forms of Digital Money: Implications for Monetary and Financial Stability, October 30, 2020.↵
163. Hofmann Christian, Which Markets Need Central Bank Digital Currency?, 18(3) *Capital Markets Law Journal* 281–302 (2023).↵
164. Hofmann Christian, The Impact of a Digital Dollar on Singapore’s Society, Banking Sector and Monetary System, 39(3) *Banking and Finance Law Review* 381–407 (2023).↵
165. Roy Davies, Electronic Money, or E-Money, and Digital Cash, February 1, 2019, <http://projects.exeter.ac.uk/RDavies/arian/emoney.html>.↵
166. Central Bank Digital Currency (CBDC), Board of Governors of the Federal Reserve System, April 11, 2023, <https://www.federalreserve.gov/cbdc-faqs.htm>.↵
167. A. Kosse and I. Mattei, Making Headway. Results of the 2022 BIS Survey on Central Bank Digital Currencies and Crypto (2023), <https://www.bis.org/publ/bppdf/bispap136.pdf>.↵
168. T. W. House, Technical Design Choices for a U.S. CBDC System (2022), <https://www.whitehouse.gov/wp-content/uploads/2022/09/09-2022-Technical-Design-Choices->

- [US-CBDC-System.pdf](#) Bank for International Settlements, Project Tourbillon—Exploring Privacy, Security and Scalability for CBDCs (2023), <https://www.bis.org/publ/othp80.htm>.↵
169. B. Brodsky, A. Dubey, and D. Tercero Lucas, Enabling Offline Payments in an Online World: A Practical Guide to Offline Payment Design (2023), <https://www.lipisadvisors.com/whitepapers>.↵
170. S. Choi, B. Kim, Y.-S. Kim, and O. Kwon, Central Bank Digital Currency and Privacy: A Randomized Survey Experiment (2023), <https://www.bis.org/publ/work1147.htm>.↵
171. Panagiotis Michalopoulos, Odunayo Olowookere, Nadia Pocher, Johannes Sedlmeir, Andreas Veneris, and Poonam Puri, Compliance Design Options for Offline CBDCs: Balancing Privacy and AML/CFT, <https://ssrn.com/abstract=4770513> p1.↵
172. Nitin Rane, Role and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Finance and Accounting (2023), <http://dx.doi.org/10.2139/ssrn.4603206> P7.↵
173. B. D. Lund and T. Wang, Chatting About ChatGPT: How May AI and GPT Impact Academia and Libraries?, 40(3) *Library Hi Tech News* 26–29 (2023).↵
174. Nitin Rane, Role and Challenges of ChatGPT and Similar Generative Artificial Intelligence in Finance and Accounting (2023), <http://dx.doi.org/10.2139/ssrn.4603206> p. 7.↵
175. Nitin Rane, Saurabh Choudhary, and Jayesh Rane, Blockchain and Artificial Intelligence (AI) Integration for Revolutionizing Security and Transparency in Finance (2023), <http://dx.doi.org/10.2139/ssrn.4644253> p1.↵
176. Nitin Rane, Saurabh Choudhary, and Jayesh Rane, Blockchain and Artificial Intelligence (AI) Integration for Revolutionizing Security and Transparency in Finance (2023), <http://dx.doi.org/10.2139/ssrn.4644253> p13.↵

177. M. Almada and N. Petit, The EU AI Act: A Medley of Product Safety and Fundamental Rights? 62(1) *Common Market Law Review* 85–120 (2025). OECD, AI Language Models: Technological, Socio-economic, and Policy Considerations (DSTI/CDEP/AIGO(2022)1/FINAL) (2023).[↩]
178. Regulation (EU) 2024/1689. EU Official Journal on 12 July 2024.[↩]
179. Art. 9 Regulation (EU) 2016/679 (GDPR).[↩]
180. Maria Lucia Passador, AI in the Vault: AI Act’s Impact on Financial Regulation. Bocconi Legal Studies Research Paper No. 4898828 (Jul. 1, 2024), <https://ssrn.com/abstract=4898828>.[↩]
181. See Art. 26(5), Art. 74(6, 11, 12, 13), Art. 85, 79, 99.[↩]
182. Marco Almada, The EU AI Act: Logic, Content, and implications for finance, in *Digital Finance in the EU: Navigating new technological trends and the AI revolution*, Editors Lorenzo Moretti, Laura Rinaldi, and Pierre Schlosser (European University Institute, 2025), 111.[↩]
183. Antonella Sciarrone Alibrandi et al., The European AI Act’s Impact on Financial Markets: From Governance to Co-Regulation (Eur. Banking Inst. Working Paper Series, Paper No. 138, 2023), <https://ssrn.com/abstract=4414559>[↩]
184. Maria Lucia Passador, AI in the Vault: AI Act’s Impact on Financial Regulation (Bocconi Legal Studies Research Paper No. 4898828, July 1, 2024), <https://ssrn.com/abstract=4898828>.[↩]
185. EU AI Act, arts. 18, 19, 72 (describing what documentation and logs financial institutions are required to keep on the high-risk AI systems).[↩]
186. Robert Kilian, Linda Jäck, and Dominik Ebel, European AI Standards—Technical Standardization and Implementation

Challenges under the EU AI Act (Feb. 26, 2025),
<https://ssrn.com/abstract=5155591> P27.↵

OceanofPDF.com